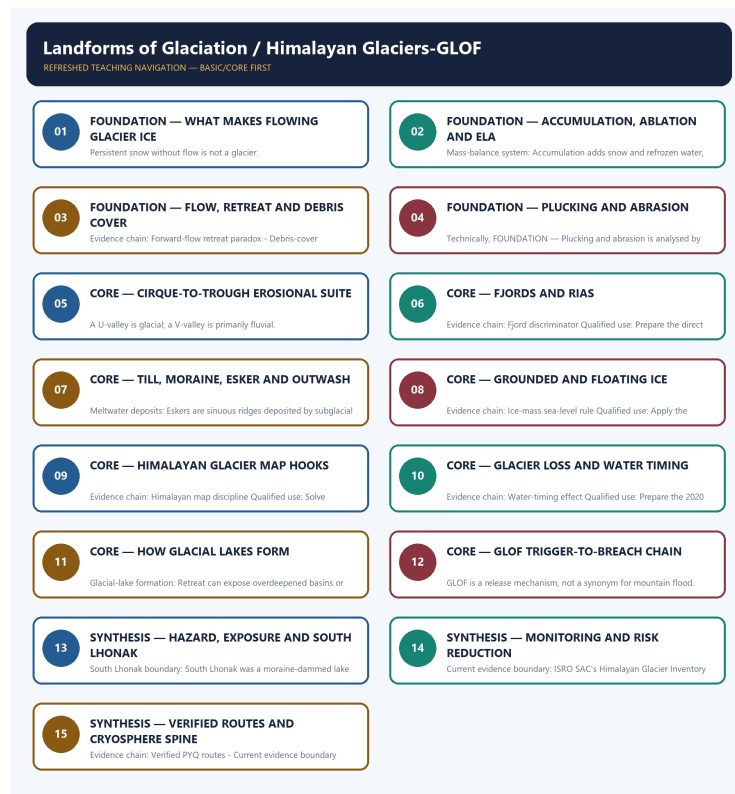


COMPLETE LEARNING SESSION

Landforms of Glaciation / Himalayan Glaciers-GLOF - Learner-v2 Refreshed

UPSC complete learning session | Foundation to advanced | Concepts, evidence, practice and answer writing



Original deterministic study visual; labels are explanatory, not textual quotations.

Source order: repository knowledge -> official current linkage -> clearly marked analysis. No fabricated PYQs or statistics.

CONTENTS / SESSION INDEX

Major learning parts and meaningful subtopics are listed in document order. Page numbers are generated from the final PDF layout; PDF bookmarks mirror the same hierarchy.

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Landforms of Glaciation / Himalayan Glaciers-GLOF - Learner-v2 Complete Learning Session

Authoring-only generation: 2026-09-01. No PDF was rendered and no tracker or index was mutated.

SOURCE, PROGRESSION AND CURRENT-LINKAGE AUDIT

- **Generation date:** 2026-09-01.
- **Repository-first evidence:** the Basic owner is taught first and preserved in full; the Advanced owner is retained only in the optional final teaching block.
- **OCR evidence:** Repository Markdown was primary. The three required local Geography books were inspected as supplementary OCR/searchable evidence. No unsupported page precision, volatile figure or quotation was imported.
- **Qdrant:** not used; repository Markdown and available OCR context were sufficient.
- **PYQ integrity:** Verified routing retains the 2020 and 2023 GS-I questions and the 2019 glacier-river objective route. Older objective keys unavailable locally remain explicitly unverified.
- **Live-link boundary:** The ISRO SAC Himalayan Glacier Inventory Atlas was directly fetched on 2026-09-01 as an official mapping portal. The CWC GLOF page timed out during the fresh check, so the package retains only the already verified process boundary and states no current rate, count, ranking or forecast.
- **Fact/inference discipline:** no current-affairs item, PYQ wording, figure or quotation is invented.

BASIC LEARNING SESSION

DEEP-REVIEW LEARNING CONTRACT

Control	Binding rule for this package
Syllabus boundary	Complete physical process, spatial pattern and India anchor are taught before optional Advanced depth.
Causal method	Initial condition -> energy/force or driver -> mechanism -> landform/climate/ocean pattern -> consequence -> feedback/limit.
Spatial method	Latitude, altitude, continentality, relief, plate setting, basin/coast orientation and map scale are explicit.
Terminology	Close terms are defined on common axes; process, form, region, hazard, regulation and current status are never collapsed.
Evidence method	Claim -> named place/map/process/official record -> analysis -> qualification.
Practice contract	Every solved item has demand decoding, a detailed examiner-grade model, executable timed/compression plan, marks rationale and answer-specific improvement.
Approval	This immutable successor remains approved: false pending explicit approval.

Canonical Basic/Core owner:

upsc-ai-kit\knowledge\Geography\basic\06_Landforms-of-Glaciation.md **Canonical topic owner:**

upsc-ai-kit\knowledge\Geography\basic\06_Landforms-of-Glaciation.md **Optional Advanced**

owner: upsc-ai-kit\knowledge\Geography\advanced\06_Himalayan-Glaciers-GLOF.md **Official**

syllabus mapping: upsc-ai-kit\knowledge\Geography\OFFICIAL-UPSC-SYLLABUS-MAPPING.md

EVIDENCE, PYQ AND CURRENT-STATUS CONTROL

- **Process discipline:** no feature is listed without formation sequence, controlling variables and a limiting condition.
- **Map discipline:** distribution is reconstructed through latitude, plate/relief setting, wind/current direction, drainage/coast orientation and named anchors.
- **Quantitative discipline:** coordinates, thresholds, magnitudes, dates, counts and project dimensions retain units, source/date and uncertainty.
- **Causal discipline:** association is not treated as sufficiency; interacting controls, scale, lag, feedback and exceptions are stated.
- **PYQ discipline:** repository routing ledgers and held papers control wording and metadata; reconstructed wording and unavailable official keys remain labelled.
- **Status discipline:** every changeable claim retains an official source, observation date and status boundary.
- **Current-status note, rechecked 2026-09-05:** Rechecked 2026-09-05: SAC's official Himalayan Glacier Inventory Atlas remains the inventory portal. ISRO's 22 April 2024 satellite assessment compares 2,431 glacial lakes larger than 10 hectares in the 2016-17 inventory with the 1984-onward record and identifies 676 expanding lakes, including 130 within India; these are inventory- and period-specific observations, not current GLOF probabilities. CWC's official GLOF explainer and monitoring portal require close monitoring of Himalayan glacial lakes, while NDMA's national guideline supplies the hazard-assessment, warning, mitigation and preparedness framework. No lake is labelled dangerous from area growth alone.

Live/official context sources rechecked for this generation:

- https://vedas.sac.gov.in/en/Himalayan_Glacier_Inventory_Atlas.html
- https://www.isro.gov.in/Satellite_Insights_Expanding_Glacial_Lakes_Indian_Himalayas.html
- <https://cwc.gov.in/glacial-lake-outburst-floods-glof>
- <https://cwc.gov.in/glacial-lakeswater-bodies-himalayan-region>
- https://mitigation.ndma.gov.in/ndmahr-admin/public/uploads/advertisement_document/file63791735554350.pdf



Refreshed teaching navigation

Distinct embedded teaching-navigation image. The separate continuous Cārvāka-style flowchart package remains an independent at-a-glance artifact.

SESSION 1 - FOUNDATION - What makes flowing glacier ice

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: What makes flowing glacier ice explains how Glacier definition shape the examinable geography of Landforms of Glaciation / Himalayan Glaciers-GLOF.

Technical definition: In geographical analysis, What makes flowing glacier ice is the source-bounded relation among Glacier definition, classified by process, form, spatial setting, temporal change and human consequence.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

What makes flowing glacier ice must be explained as a process-to-pattern chain, not as an unconnected catalogue of landforms, places or schemes.

MUST-WRITE KEYWORDS

- What
- makes
- flowing
- glacier
- definition

How to use them: Define the process, locate its spatial expression, cite named evidence, apply this limit - Persistent snow without flow is not a glacier. - and conclude: Define the land-ice system before naming forms.

VISUAL FIRST

WHAT MAKES FLOWING GLACIER ICE

01. Glacier definition

BOUNDARY -> Persistent snow without flow is not a glacier.

This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.

CORE EXPLANATION

A glacier is compacted and recrystallised land ice thick enough to deform or slide under gravity; a persistent snowfield need not be a glacier because it may not flow.

NAMED EVIDENCE AND MECHANISM

- **Glacier definition:** A glacier is compacted and recrystallised land ice thick enough to deform or slide under gravity; a persistent snowfield need not be a glacier because it may not flow.

EXAMINER CAUTION

- Persistent snow without flow is not a glacier.

EXAM LINK

- **Prelims:** Separate the agent, landform, location, status and scale attached to What makes flowing glacier ice.
- **Mains:** Define the land-ice system before naming forms.

MINI RECAP

- **Evidence chain:** Glacier definition
- **Qualified use:** Define the land-ice system before naming forms.

CLOSING RECALL FLOW

SUBTOPIC CLOSURE FLOW | SUBTOPIC CLOSURE FLOW

1. KEY TERMS / DEFINITIONS

What | makes | flowing | glacier | definition

2. MECHANISM / ARGUMENT

connect Glacier definition through process, place and time

3. CONSEQUENCE / CONTRAST

Define the land-ice system before naming forms.

4. UPSC TRAP / ANSWER-USE

Persistent snow without flow is not a glacier.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: What makes flowing glacier ice converts physical process into a qualified spatial argument

SESSION 2 - FOUNDATION - Accumulation, ablation and ELA

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Accumulation, ablation and ELA explains how Mass-balance system and ELA, snowline and snout shape the examinable geography of Landforms of Glaciation / Himalayan Glaciers-GLOF.

Technical definition: In geographical analysis, Accumulation, ablation and ELA is the source-bounded relation among Mass-balance system and ELA, snowline and snout, classified by process, form, spatial setting, temporal change and human consequence.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

Accumulation, ablation and ELA must be explained as a process-to-pattern chain, not as an unconnected catalogue of landforms, places or schemes.

MUST-WRITE KEYWORDS

- Accumulation
- ablation
- Mass-balance
- snowline
- snout

How to use them: Define the process, locate its spatial expression, cite named evidence, apply this limit - Every balance claim needs a glacier and period. - and conclude: Use budget vocabulary instead of temperature slogans.

VISUAL FIRST

ACCUMULATION, ABLATION AND ELA

01. Mass-balance system

|
v

02. ELA, snowline and snout

BOUNDARY -> Every balance claim needs a glacier and period.

This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.

CORE EXPLANATION

Accumulation adds snow and refrozen water, ablation removes mass, and net balance is their difference over a stated glacier and period. The equilibrium-line altitude balances annual accumulation and ablation; the climatic snowline is a long-term regional limit, while the snout is the glacier terminus.

NAMED EVIDENCE AND MECHANISM

- **Mass-balance system:** Accumulation adds snow and refrozen water, ablation removes mass, and net balance is their difference over a stated glacier and period.
- **ELA, snowline and snout:** The equilibrium-line altitude balances annual accumulation and ablation; the climatic snowline is a long-term regional limit, while the snout is the glacier terminus.

EXAMINER CAUTION

- Every balance claim needs a glacier and period.

EXAM LINK

- **Prelims:** Separate the agent, landform, location, status and scale attached to Accumulation, ablation and ELA.
- **Mains:** Use budget vocabulary instead of temperature slogans.

MINI RECAP

- **Evidence chain:** Mass-balance system -> ELA, snowline and snout
- **Qualified use:** Use budget vocabulary instead of temperature slogans.

CLOSING RECALL FLOW

SUBTOPIC CLOSURE FLOW | SUBTOPIC CLOSURE FLOW

1. KEY TERMS / DEFINITIONS

Accumulation | ablation | Mass-balance | snowline | snout

2. MECHANISM / ARGUMENT

connect Mass-balance system and ELA, snowline and snout through process, place and time

3. CONSEQUENCE / CONTRAST

Use budget vocabulary instead of temperature slogans.

4. UPSC TRAP / ANSWER-USE

Every balance claim needs a glacier and period.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Accumulation, ablation and ELA converts physical process into a qualified spatial argument

SESSION 3 - FOUNDATION - Flow, retreat and debris cover

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Flow, retreat and debris cover explains how Forward-flow retreat paradox and Debris-cover qualification shape the examinable geography of Landforms of Glaciation / Himalayan Glaciers-GLOF.

Technical definition: In geographical analysis, Flow, retreat and debris cover is the source-bounded relation among Forward-flow retreat paradox and Debris-cover qualification, classified by process, form, spatial setting, temporal change and human consequence.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

Flow, retreat and debris cover must be explained as a process-to-pattern chain, not as an unconnected catalogue of landforms, places or schemes.

MUST-WRITE KEYWORDS

- Flow
- retreat
- debris
- cover
- Forward-flow
- paradox

How to use them: Define the process, locate its spatial expression, cite named evidence, apply this limit - Snout movement is not a complete mass-balance metric. - and conclude: Explain geometry through delivery versus loss.

VISUAL FIRST

FLOW, RETREAT AND DEBRIS COVER

01. Forward-flow retreat paradox

|
v

02. Debris-cover qualification

BOUNDARY -> Snout movement is not a complete mass-balance metric.

This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.

CORE EXPLANATION

Ice can continue moving down-valley while the snout retreats when ablation at the terminus exceeds forward ice delivery. Thin debris can enhance melt by lowering albedo, whereas sufficiently thick debris can insulate; debris-covered tongues may thin without rapid visible snout retreat.

NAMED EVIDENCE AND MECHANISM

- **Forward-flow retreat paradox:** Ice can continue moving down-valley while the snout retreats when ablation at the terminus exceeds forward ice delivery.
- **Debris-cover qualification:** Thin debris can enhance melt by lowering albedo, whereas sufficiently thick debris can insulate; debris-covered tongues may thin without rapid visible snout retreat.

EXAMINER CAUTION

- Snout movement is not a complete mass-balance metric.

EXAM LINK

- **Prelims:** Separate the agent, landform, location, status and scale attached to Flow, retreat and debris cover.
- **Mains:** Explain geometry through delivery versus loss.

MINI RECAP

- **Evidence chain:** Forward-flow retreat paradox -> Debris-cover qualification
- **Qualified use:** Explain geometry through delivery versus loss.

CLOSING RECALL FLOW

SUBTOPIC CLOSURE FLOW SUBTOPIC CLOSURE FLOW			
1. KEY TERMS / DEFINITIONS Flow retreat debris cover Forward-flow paradox	2. MECHANISM / ARGUMENT connect Forward-flow retreat paradox and Debris-cover qualification through process, place and time	3. CONSEQUENCE / CONTRAST Explain geometry through delivery versus loss.	4. UPSC TRAP / ANSWER-USE Snout movement is not a complete mass-balance metric.
ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Flow, retreat and debris cover converts physical process into a qualified spatial argument			

SESSION 4 - FOUNDATION - Plucking and abrasion

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Plucking and abrasion explains how Glacial erosion shape the examinable geography of Landforms of Glaciation / Himalayan Glaciers-GLOF.

Technical definition: In geographical analysis, Plucking and abrasion is the source-bounded relation among Glacial erosion, classified by process, form, spatial setting, temporal change and human consequence.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

Plucking and abrasion must be explained as a process-to-pattern chain, not as an unconnected catalogue of landforms, places or schemes.

MUST-WRITE KEYWORDS

- **Plucking**
- **abrasion**
- **Glacial**
- **erosion**

How to use them: Define the process, locate its spatial expression, cite named evidence, apply this limit - Both processes can operate together on structured bedrock. - and conclude: Link mechanism to polish, striation and asymmetry.

VISUAL FIRST

PLUCKING AND ABRASION

01. Glacial erosion

BOUNDARY -> Both processes can operate together on structured bedrock.

This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.

CORE EXPLANATION

Plucking detaches jointed blocks and abrasion by entrained debris polishes, striates and grinds bedrock, producing rock flour and asymmetric roche moutonnee forms.

NAMED EVIDENCE AND MECHANISM

- **Glacial erosion:** Plucking detaches jointed blocks and abrasion by entrained debris polishes, striates and grinds bedrock, producing rock flour and asymmetric roche moutonnee forms.

EXAMINER CAUTION

- Both processes can operate together on structured bedrock.

EXAM LINK

- **Prelims:** Separate the agent, landform, location, status and scale attached to Plucking and abrasion.
- **Mains:** Link mechanism to polish, striation and asymmetry.

MINI RECAP

- **Evidence chain:** Glacial erosion
- **Qualified use:** Link mechanism to polish, striation and asymmetry.

CLOSING RECALL FLOW

SUBTOPIC CLOSURE FLOW SUBTOPIC CLOSURE FLOW			
1. KEY TERMS / DEFINITIONS Plucking abrasion Glacial erosion	2. MECHANISM / ARGUMENT connect Glacial erosion through process, place and time	3. CONSEQUENCE / CONTRAST Link mechanism to polish, striation and asymmetry.	4. UPSC TRAP / ANSWER-USE Both processes can operate together on structured bedrock.
ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Plucking and abrasion converts physical process into a qualified spatial argument			

SESSION 5 - CORE - Cirque-to-trough erosional suite

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Cirque-to-trough erosional suite explains how Erosional landform suite shape the examinable geography of Landforms of Glaciation / Himalayan Glaciers-GLOF.

Technical definition: In geographical analysis, Cirque-to-trough erosional suite is the source-bounded relation among Erosional landform suite, classified by process, form, spatial setting, temporal change and human consequence.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

Cirque-to-trough erosional suite must be explained as a process-to-pattern chain, not as an unconnected catalogue of landforms, places or schemes.

MUST-WRITE KEYWORDS

- **Cirque-to-trough**
- **erosional**
- **suite**
- **landform**

How to use them: Define the process, locate its spatial expression, cite named evidence, apply this limit - A U-valley is glacial; a V-valley is primarily fluvial. - and conclude: Reconstruct the erosional landscape.

VISUAL FIRST

CIRQUE-TO-TROUGH EROSIONAL SUITE

01. Erosional landform suite

BOUNDARY -> A U-valley is glacial; a V-valley is primarily fluvial.

This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.

CORE EXPLANATION

Cirques, aretes and horns develop around glacier heads; valley ice widens and deepens V-valleys into U-shaped troughs and leaves hanging tributary valleys.

NAMED EVIDENCE AND MECHANISM

- **Erosional landform suite:** Cirques, aretes and horns develop around glacier heads; valley ice widens and deepens V-valleys into U-shaped troughs and leaves hanging tributary valleys.

EXAMINER CAUTION

- A U-valley is glacial; a V-valley is primarily fluvial.

EXAM LINK

- **Prelims:** Separate the agent, landform, location, status and scale attached to Cirque-to-trough erosional suite.
- **Mains:** Reconstruct the erosional landscape.

MINI RECAP

- **Evidence chain:** Erosional landform suite
- **Qualified use:** Reconstruct the erosional landscape.

CLOSING RECALL FLOW

SUBTOPIC CLOSURE FLOW SUBTOPIC CLOSURE FLOW			
1. KEY TERMS / DEFINITIONS Cirque-to-trough erosional suite landform	2. MECHANISM / ARGUMENT connect Erosional landform suite through process, place and time	3. CONSEQUENCE / CONTRAST Reconstruct the erosional landscape.	4. UPSC TRAP / ANSWER-USE A U-valley is glacial; a V-valley is primarily fluvial.
ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Cirque-to-trough erosional suite converts physical process into a qualified spatial argument			

SESSION 6 - CORE - Fjords and rias

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Fjords and rias explains how Fjord discriminator shape the examinable geography of Landforms of Glaciation / Himalayan Glaciers-GLOF.

Technical definition: In geographical analysis, Fjords and rias is the source-bounded relation among Fjord discriminator, classified by process, form, spatial setting, temporal change and human consequence.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

Fjords and rias must be explained as a process-to-pattern chain, not as an unconnected catalogue of landforms, places or schemes.

MUST-WRITE KEYWORDS

- Fjords
- rias
- Fjord
- discriminator

How to use them: Define the process, locate its spatial expression, cite named evidence, apply this limit - The sea floods an existing trough; it does not excavate it. - and conclude: Prepare the direct 2023 route.

VISUAL FIRST

FJORDS AND RIAS

01. Fjord discriminator

BOUNDARY -> The sea floods an existing trough; it does not excavate it.

This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.

CORE EXPLANATION

A fjord is a marine-flooded glacial trough, whereas a ria is a drowned river valley; marine inundation follows rather than creates the glacial excavation.

NAMED EVIDENCE AND MECHANISM

- **Fjord discriminator:** A fjord is a marine-flooded glacial trough, whereas a ria is a drowned river valley; marine inundation follows rather than creates the glacial excavation.

EXAMINER CAUTION

- The sea floods an existing trough; it does not excavate it.

EXAM LINK

- **Prelims:** Separate the agent, landform, location, status and scale attached to Fjords and rias.
- **Mains:** Prepare the direct 2023 route.

MINI RECAP

- **Evidence chain:** Fjord discriminator
- **Qualified use:** Prepare the direct 2023 route.

CLOSING RECALL FLOW

SUBTOPIC CLOSURE FLOW SUBTOPIC CLOSURE FLOW			
1. KEY TERMS / DEFINITIONS Fjords rias Fjord discriminator	2. MECHANISM / ARGUMENT connect Fjord discriminator through process, place and time	3. CONSEQUENCE / CONTRAST Prepare the direct 2023 route.	4. UPSC TRAP / ANSWER-USE The sea floods an existing trough; it does not excavate it.
ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Fjords and rias converts physical process into a qualified spatial argument			

SESSION 7 - CORE - Till, moraine, esker and outwash

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Till, moraine, esker and outwash explains how Till and moraine and Meltwater deposits shape the examinable geography of Landforms of Glaciation / Himalayan Glaciers-GLOF.

Technical definition: In geographical analysis, Till, moraine, esker and outwash is the source-bounded relation among Till and moraine and Meltwater deposits, classified by process, form, spatial setting, temporal change and human consequence.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

Till, moraine, esker and outwash must be explained as a process-to-pattern chain, not as an unconnected catalogue of landforms, places or schemes.

MUST-WRITE KEYWORDS

- Till
- moraine
- esker
- outwash
- Meltwater
- deposits

How to use them: Define the process, locate its spatial expression, cite named evidence, apply this limit - Sorting reveals whether ice or meltwater deposited the load. - and conclude: Use agent and texture for elimination.

VISUAL FIRST

TILL, MORaine, ESKER AND OUTWASH

01. Till and moraine

|
v

02. Meltwater deposits

BOUNDARY -> Sorting reveals whether ice or meltwater deposited the load.

This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.

CORE EXPLANATION

Direct ice deposition is generally unsorted till; lateral, medial, terminal and ground moraines describe debris position or form. Eskers are sinuous ridges deposited by subglacial or englacial streams, while outwash plains contain relatively sorted sediment spread by braided meltwater.

NAMED EVIDENCE AND MECHANISM

- **Till and moraine:** Direct ice deposition is generally unsorted till; lateral, medial, terminal and ground moraines describe debris position or form.
- **Meltwater deposits:** Eskers are sinuous ridges deposited by subglacial or englacial streams, while outwash plains contain relatively sorted sediment spread by braided meltwater.

EXAMINER CAUTION

- Sorting reveals whether ice or meltwater deposited the load.

EXAM LINK

- **Prelims:** Separate the agent, landform, location, status and scale attached to Till, moraine, esker and outwash.
- **Mains:** Use agent and texture for elimination.

MINI RECAP

- **Evidence chain:** Till and moraine -> Meltwater deposits
- **Qualified use:** Use agent and texture for elimination.

CLOSING RECALL FLOW

SUBTOPIC CLOSURE FLOW SUBTOPIC CLOSURE FLOW			
1. KEY TERMS / DEFINITIONS Till moraine esker outwash Meltwater deposits	2. MECHANISM / ARGUMENT connect Till and moraine and Meltwater deposits through process, place and time	3. CONSEQUENCE / CONTRAST Use agent and texture for elimination.	4. UPSC TRAP / ANSWER-USE Sorting reveals whether ice or meltwater deposited the load.
ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Till, moraine, esker and outwash converts physical process into a qualified spatial argument			

SESSION 8 - CORE - Grounded and floating ice

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Grounded and floating ice explains how Ice-mass sea-level rule shape the examinable geography of Landforms of Glaciation / Himalayan Glaciers-GLOF.

Technical definition: In geographical analysis, Grounded and floating ice is the source-bounded relation among Ice-mass sea-level rule, classified by process, form, spatial setting, temporal change and human consequence.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

Grounded and floating ice must be explained as a process-to-pattern chain, not as an unconnected catalogue of landforms, places or schemes.

MUST-WRITE KEYWORDS

- Grounded
- floating
- Ice-mass
- sea-level
- rule

How to use them: Define the process, locate its spatial expression, cite named evidence, apply this limit - Ice shelf loss can matter indirectly through buttressing. - and conclude: Apply the displacement rule carefully.

VISUAL FIRST

GROUNDING AND FLOATING ICE

01. Ice-mass sea-level rule

BOUNDARY -> Ice shelf loss can matter indirectly through buttressing.

This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.

CORE EXPLANATION

Loss of grounded glaciers, ice caps and ice sheets adds ocean mass; floating sea ice or shelf melt does not directly add mass beyond displacement, though shelves can buttress grounded ice.

NAMED EVIDENCE AND MECHANISM

- **Ice-mass sea-level rule:** Loss of grounded glaciers, ice caps and ice sheets adds ocean mass; floating sea ice or shelf melt does not directly add mass beyond displacement, though shelves can buttress grounded ice.

EXAMINER CAUTION

- Ice shelf loss can matter indirectly through buttressing.

EXAM LINK

- **Prelims:** Separate the agent, landform, location, status and scale attached to Grounded and floating ice.
- **Mains:** Apply the displacement rule carefully.

MINI RECAP

- **Evidence chain:** Ice-mass sea-level rule
- **Qualified use:** Apply the displacement rule carefully.

CLOSING RECALL FLOW

SUBTOPIC CLOSURE FLOW SUBTOPIC CLOSURE FLOW			
1. KEY TERMS / DEFINITIONS Grounded floating Ice-mass sea-level rule	2. MECHANISM / ARGUMENT connect Ice-mass sea-level rule through process, place and time	3. CONSEQUENCE / CONTRAST Apply the displacement rule carefully.	4. UPSC TRAP / ANSWER-USE Ice shelf loss can matter indirectly through buttressing.
ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Grounded and floating ice converts physical process into a qualified spatial argument			

SESSION 9 - CORE - Himalayan glacier map hooks

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Himalayan glacier map hooks explains how Himalayan map discipline shape the examinable geography of Landforms of Glaciation / Himalayan Glaciers-GLOF.

Technical definition: In geographical analysis, Himalayan glacier map hooks is the source-bounded relation among Himalayan map discipline, classified by process, form, spatial setting, temporal change and human consequence.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

Himalayan glacier map hooks must be explained as a process-to-pattern chain, not as an unconnected catalogue of landforms, places or schemes.

MUST-WRITE KEYWORDS

- Himalayan
- glacier
- hooks
- discipline

How to use them: Define the process, locate its spatial expression, cite named evidence, apply this limit - Physical mapping must not be turned into a political-boundary assertion. - and conclude: Solve glacier-river matching.

VISUAL FIRST

HIMALAYAN GLACIER MAP HOOKS

01. Himalayan map discipline

BOUNDARY -> Physical mapping must not be turned into a political-boundary assertion.

This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.

CORE EXPLANATION

Gangotri links to the Bhagirathi, Zemu to the Teesta, Bara Shigri to the Chenab system and Siachen to the Nubra-Shyok-Indus system; map claims must remain distinct from boundary claims.

NAMED EVIDENCE AND MECHANISM

- **Himalayan map discipline:** Gangotri links to the Bhagirathi, Zemu to the Teesta, Bara Shigri to the Chenab system and Siachen to the Nubra-Shyok-Indus system; map claims must remain distinct from boundary claims.

EXAMINER CAUTION

- Physical mapping must not be turned into a political-boundary assertion.

EXAM LINK

- **Prelims:** Separate the agent, landform, location, status and scale attached to Himalayan glacier map hooks.
- **Mains:** Solve glacier-river matching.

MINI RECAP

- **Evidence chain:** Himalayan map discipline
- **Qualified use:** Solve glacier-river matching.

CLOSING RECALL FLOW

SUBTOPIC CLOSURE FLOW SUBTOPIC CLOSURE FLOW			
1. KEY TERMS / DEFINITIONS Himalayan glacier hooks discipline	2. MECHANISM / ARGUMENT connect Himalayan map discipline through process, place and time	3. CONSEQUENCE / CONTRAST Solve glacier-river matching.	4. UPSC TRAP / ANSWER-USE Physical mapping must not be turned into a political-boundary assertion.
ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Himalayan glacier map hooks converts physical process into a qualified spatial argument			

SESSION 10 - CORE - Glacier loss and water timing

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Glacier loss and water timing explains how Water-timing effect shape the examinable geography of Landforms of Glaciation / Himalayan Glaciers-GLOF.

Technical definition: In geographical analysis, Glacier loss and water timing is the source-bounded relation among Water-timing effect, classified by process, form, spatial setting, temporal change and human consequence.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

Glacier loss and water timing must be explained as a process-to-pattern chain, not as an unconnected catalogue of landforms, places or schemes.

MUST-WRITE KEYWORDS

- **Glacier**
- **loss**
- **water**
- **timing**
- **Water-timing**
- **effect**

How to use them: Define the process, locate its spatial expression, cite named evidence, apply this limit - Avoid uniform annual-flow-collapse claims. - and conclude: Prepare the 2020 GS-I water-resources answer.

VISUAL FIRST

GLACIER LOSS AND WATER TIMING

01. Water-timing effect

BOUNDARY -> Avoid uniform annual-flow-collapse claims.

This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.

CORE EXPLANATION

Sustained ice loss can initially raise local melt contribution but later weaken lean-season and drought-year buffering; monsoon rain, snow and groundwater prevent a uniform all-rivers-dry conclusion.

NAMED EVIDENCE AND MECHANISM

- **Water-timing effect:** Sustained ice loss can initially raise local melt contribution but later weaken lean-season and drought-year buffering; monsoon rain, snow and groundwater prevent a uniform all-rivers-dry conclusion.

EXAMINER CAUTION

- Avoid uniform annual-flow-collapse claims.

EXAM LINK

- **Prelims:** Separate the agent, landform, location, status and scale attached to Glacier loss and water timing.
- **Mains:** Prepare the 2020 GS-I water-resources answer.

MINI RECAP

- **Evidence chain:** Water-timing effect
- **Qualified use:** Prepare the 2020 GS-I water-resources answer.

CLOSING RECALL FLOW

SUBTOPIC CLOSURE FLOW SUBTOPIC CLOSURE FLOW			
1. KEY TERMS / DEFINITIONS Glacier loss water timing Water-timing effect	2. MECHANISM / ARGUMENT connect Water-timing effect through process, place and time	3. CONSEQUENCE / CONTRAST Prepare the 2020 GS-I water-resources answer.	4. UPSC TRAP / ANSWER-USE Avoid uniform annual-flow-collapse claims.
ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Glacier loss and water timing converts physical process into a qualified spatial argument			

SESSION 11 - CORE - How glacial lakes form

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: How glacial lakes form explains how Glacial-lake formation shape the examinable geography of Landforms of Glaciation / Himalayan Glaciers-GLOF.

Technical definition: In geographical analysis, How glacial lakes form is the source-bounded relation among Glacial-lake formation, classified by process, form, spatial setting, temporal change and human consequence.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

How glacial lakes form must be explained as a process-to-pattern chain, not as an unconnected catalogue of landforms, places or schemes.

MUST-WRITE KEYWORDS

- glacial
- lakes
- form
- Glacial-lake
- formation

How to use them: Define the process, locate its spatial expression, cite named evidence, apply this limit - Lake growth is not identical to dam instability. - and conclude: Separate inventory from hazard assessment.

VISUAL FIRST

HOW GLACIAL LAKES FORM

01. Glacial-lake formation

BOUNDARY -> Lake growth is not identical to dam instability.

This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.

CORE EXPLANATION

Retreat can expose overdeepened basins or leave moraine dams that store water; lake existence alone does not establish instability or downstream risk.

NAMED EVIDENCE AND MECHANISM

- **Glacial-lake formation:** Retreat can expose overdeepened basins or leave moraine dams that store water; lake existence alone does not establish instability or downstream risk.

EXAMINER CAUTION

- Lake growth is not identical to dam instability.

EXAM LINK

- **Prelims:** Separate the agent, landform, location, status and scale attached to How glacial lakes form.
- **Mains:** Separate inventory from hazard assessment.

MINI RECAP

- **Evidence chain:** Glacial-lake formation
- **Qualified use:** Separate inventory from hazard assessment.

CLOSING RECALL FLOW

SUBTOPIC CLOSURE FLOW SUBTOPIC CLOSURE FLOW			
1. KEY TERMS / DEFINITIONS glacial lakes form Glacial-lake formation	2. MECHANISM / ARGUMENT connect Glacial-lake formation through process, place and time	3. CONSEQUENCE / CONTRAST Separate inventory from hazard assessment.	4. UPSC TRAP / ANSWER-USE Lake growth is not identical to dam instability.
ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: How glacial lakes form converts physical process into a qualified spatial argument			

SESSION 12 - CORE - GLOF trigger-to-breach chain

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: GLOF trigger-to-breach chain explains how GLOF mechanism shape the examinable geography of Landforms of Glaciation / Himalayan Glaciers-GLOF.

Technical definition: In geographical analysis, GLOF trigger-to-breach chain is the source-bounded relation among GLOF mechanism, classified by process, form, spatial setting, temporal change and human consequence.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

GLOF trigger-to-breach chain must be explained as a process-to-pattern chain, not as an unconnected catalogue of landforms, places or schemes.

MUST-WRITE KEYWORDS

- GLOF
- trigger-to-breach
- chain
- classification
- causation
- comparison

How to use them: Define the process, locate its spatial expression, cite named evidence, apply this limit - GLOF is a release mechanism, not a synonym for mountain flood. - and conclude: Explain trigger, overtopping or failure and cascade.

VISUAL FIRST

GLOF TRIGGER-TO-BREACH CHAIN

01. GLOF mechanism

BOUNDARY -> GLOF is a release mechanism, not a synonym for mountain flood.

This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.

CORE EXPLANATION

A GLOF is sudden lake release after overtopping or dam failure; triggers can include slope failure, ice or rock entry, intense inflow, piping, erosion or seismic disturbance.

NAMED EVIDENCE AND MECHANISM

- **GLOF mechanism:** A GLOF is sudden lake release after overtopping or dam failure; triggers can include slope failure, ice or rock entry, intense inflow, piping, erosion or seismic disturbance.

EXAMINER CAUTION

- GLOF is a release mechanism, not a synonym for mountain flood.

EXAM LINK

- **Prelims:** Separate the agent, landform, location, status and scale attached to GLOF trigger-to-breach chain.
- **Mains:** Explain trigger, overtopping or failure and cascade.

MINI RECAP

- **Evidence chain:** GLOF mechanism
- **Qualified use:** Explain trigger, overtopping or failure and cascade.

CLOSING RECALL FLOW

SUBTOPIC CLOSURE FLOW | SUBTOPIC CLOSURE FLOW

1. KEY TERMS / DEFINITIONS

GLOF | trigger-to-breach | chain | classification | causation | comparison

2. MECHANISM / ARGUMENT

connect GLOF mechanism through process, place and time

3. CONSEQUENCE / CONTRAST

Explain trigger, overtopping or failure and cascade.

4. UPSC TRAP / ANSWER-USE

GLOF is a release mechanism, not a synonym for mountain flood.

SESSION 13 - SYNTHESIS - Hazard, exposure and South Lhonak

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Hazard, exposure and South Lhonak explains how Hazard-risk distinction and South Lhonak boundary shape the examinable geography of Landforms of Glaciation / Himalayan Glaciers-GLOF.

Technical definition: In geographical analysis, Hazard, exposure and South Lhonak is the source-bounded relation among Hazard-risk distinction and South Lhonak boundary, classified by process, form, spatial setting, temporal change and human consequence.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

Hazard, exposure and South Lhonak must be explained as a process-to-pattern chain, not as an unconnected catalogue of landforms, places or schemes.

MUST-WRITE KEYWORDS

- Hazard
- exposure
- South
- Lhonak
- Hazard-risk
- distinction

How to use them: Define the process, locate its spatial expression, cite named evidence, apply this limit - Event attribution must remain evidence-specific. - and conclude: Build a bounded Teesta case.

VISUAL FIRST

HAZARD, EXPOSURE AND SOUTH LHONAK

01. Hazard-risk distinction

|
v

02. South Lhonak boundary

BOUNDARY -> Event attribution must remain evidence-specific.

This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.

CORE EXPLANATION

GLOF hazard depends on lake, dam, slope and breach processes; risk additionally requires exposed people, settlements, roads, hydropower and limited warning or evacuation capacity. South Lhonak was a moraine-dammed lake involved in the October 2023 Teesta disaster; event-specific work examines slope movement, impulse, overtopping and breach without reducing attribution to warming alone.

NAMED EVIDENCE AND MECHANISM

- **Hazard-risk distinction:** GLOF hazard depends on lake, dam, slope and breach processes; risk additionally requires exposed people, settlements, roads, hydropower and limited warning or evacuation capacity.
- **South Lhonak boundary:** South Lhonak was a moraine-dammed lake involved in the October 2023 Teesta disaster; event-specific work examines slope movement, impulse, overtopping and breach without reducing attribution to warming alone.

EXAMINER CAUTION

- Event attribution must remain evidence-specific.

EXAM LINK

- **Prelims:** Separate the agent, landform, location, status and scale attached to Hazard, exposure and South Lhonak.

- **Mains:** Build a bounded Teesta case.

MINI RECAP

- **Evidence chain:** Hazard-risk distinction -> South Lhonak boundary
- **Qualified use:** Build a bounded Teesta case.

CLOSING RECALL FLOW

SUBTOPIC CLOSURE FLOW SUBTOPIC CLOSURE FLOW			
1. KEY TERMS / DEFINITIONS Hazard exposure South Lhonak Hazard-risk distinction	2. MECHANISM / ARGUMENT connect Hazard-risk distinction and South Lhonak boundary through process, place and time	3. CONSEQUENCE / CONTRAST Build a bounded Teesta case.	4. UPSC TRAP / ANSWER-USE Event attribution must remain evidence-specific.
ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Hazard, exposure and South Lhonak converts physical process into a qualified spatial argument			

SESSION 14 - SYNTHESIS - Monitoring and risk reduction

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Monitoring and risk reduction explains how Risk-reduction stack and Current evidence boundary shape the examinable geography of Landforms of Glaciation / Himalayan Glaciers-GLOF.

Technical definition: In geographical analysis, Monitoring and risk reduction is the source-bounded relation among Risk-reduction stack and Current evidence boundary, classified by process, form, spatial setting, temporal change and human consequence.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

Monitoring and risk reduction must be explained as a process-to-pattern chain, not as an unconnected catalogue of landforms, places or schemes.

MUST-WRITE KEYWORDS

- **Monitoring**
- **risk**
- **reduction**
- **Risk-reduction**
- **stack**

How to use them: Define the process, locate its spatial expression, cite named evidence, apply this limit - Remote sensing cannot replace field assessment and last-mile warning. - and conclude: Create an end-to-end governance stack.

VISUAL FIRST

MONITORING AND RISK REDUCTION

01. Risk-reduction stack

|
v

02. Current evidence boundary

BOUNDARY -> Remote sensing cannot replace field assessment and last-mile warning.

This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.

CORE EXPLANATION

Inventory and remote sensing must lead to field assessment, thresholds and warning, communication and drills, land-use controls, evacuation routes and GLOF-informed infrastructure design. SAC's official Himalayan Glacier Inventory Atlas remains the mapping portal. ISRO's 22 April 2024 assessment compared 2,431 lakes larger than 10 hectares in the 2016-17 inventory with observations since 1984 and identified 676 expanding lakes, including 130 within India; the dated inventory does not itself establish present GLOF probability.

NAMED EVIDENCE AND MECHANISM

- **Risk-reduction stack:** Inventory and remote sensing must lead to field assessment, thresholds and warning, communication and drills, land-use controls, evacuation routes and GLOF-informed infrastructure design.
- **Current evidence boundary:** SAC's official Himalayan Glacier Inventory Atlas remains the mapping portal. ISRO's 22 April 2024 assessment compared 2,431 lakes larger than 10 hectares in the 2016-17 inventory with observations since 1984 and identified 676 expanding lakes, including 130 within India; the dated inventory does not itself establish present GLOF probability.

EXAMINER CAUTION

- Remote sensing cannot replace field assessment and last-mile warning.

EXAM LINK

- **Prelims:** Separate the agent, landform, location, status and scale attached to Monitoring and risk reduction.
- **Mains:** Create an end-to-end governance stack.

MINI RECAP

- **Evidence chain:** Risk-reduction stack -> Current evidence boundary
- **Qualified use:** Create an end-to-end governance stack.

CLOSING RECALL FLOW

SUBTOPIC CLOSURE FLOW SUBTOPIC CLOSURE FLOW			
1. KEY TERMS / DEFINITIONS Monitoring risk reduction Risk-reduction stack	2. MECHANISM / ARGUMENT connect Risk-reduction stack and Current evidence boundary through process, place and time	3. CONSEQUENCE / CONTRAST Create an end-to-end governance stack.	4. UPSC TRAP / ANSWER-USE Remote sensing cannot replace field assessment and last-mile warning.
ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Monitoring and risk reduction converts physical process into a qualified spatial argument			

SESSION 15 - SYNTHESIS - Verified routes and cryosphere spine

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Verified routes and cryosphere spine explains how Verified PYQ routes and Current evidence boundary shape the examinable geography of Landforms of Glaciation / Himalayan Glaciers-GLOF.

Technical definition: In geographical analysis, Verified routes and cryosphere spine is the source-bounded relation among Verified PYQ routes and Current evidence boundary, classified by process, form, spatial setting, temporal change and human consequence.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

Verified routes and cryosphere spine must be explained as a process-to-pattern chain, not as an unconnected catalogue of landforms, places or schemes.

MUST-WRITE KEYWORDS

- Verified
- routes
- cryosphere
- spine

How to use them: Define the process, locate its spatial expression, cite named evidence, apply this limit - Do not invent official objective keys or dynamic rates. - and conclude: Close with process, map, risk and qualification.

VISUAL FIRST

VERIFIED ROUTES AND CRYOSPHERE SPINE

01. Verified PYQ routes

|

v

02. Current evidence boundary

BOUNDARY -> Do not invent official objective keys or dynamic rates.

This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.

CORE EXPLANATION

Direct routed demands include 2020 GS-I on glacier melting and Indian water resources, 2023 GS-I on fjords and 2019 Prelims glacier-river matching. SAC's official Himalayan Glacier Inventory Atlas remains the mapping portal. ISRO's 22 April 2024 assessment compared 2,431 lakes larger than 10 hectares in the 2016-17 inventory with observations since 1984 and identified 676 expanding lakes, including 130 within India; the dated inventory does not itself establish present GLOF probability.

NAMED EVIDENCE AND MECHANISM

- **Verified PYQ routes:** Direct routed demands include 2020 GS-I on glacier melting and Indian water resources, 2023 GS-I on fjords and 2019 Prelims glacier-river matching.
- **Current evidence boundary:** SAC's official Himalayan Glacier Inventory Atlas remains the mapping portal. ISRO's 22 April 2024 assessment compared 2,431 lakes larger than 10 hectares in the 2016-17 inventory with observations since 1984 and identified 676 expanding lakes, including 130 within India; the dated inventory does not itself establish present GLOF probability.

EXAMINER CAUTION

- Do not invent official objective keys or dynamic rates.

EXAM LINK

- **Prelims:** Separate the agent, landform, location, status and scale attached to Verified routes and cryosphere spine.
- **Mains:** Close with process, map, risk and qualification.

MINI RECAP

- **Evidence chain:** Verified PYQ routes -> Current evidence boundary
- **Qualified use:** Close with process, map, risk and qualification.

CLOSING RECALL FLOW

SUBTOPIC CLOSURE FLOW | SUBTOPIC CLOSURE FLOW

1. KEY TERMS / DEFINITIONS

Verified | routes | cryosphere | spine

2. MECHANISM / ARGUMENT

connect Verified PYQ routes and Current evidence boundary through process, place and time

3. CONSEQUENCE / CONTRAST

Close with process, map, risk and qualification.

4. UPSC TRAP / ANSWER-USE

Do not invent official objective keys or dynamic rates.

COMPLETE BASIC OWNER EVIDENCE BANK

Subject: Geography · **Tier:** Must-Do (foundation + standard) · **GS Paper:** GS-I **Grounded in:** G.C. Leong, *Certificate Physical & Human Geography* + Majid Husain, *Indian & World Geography*. **FACT:** = from source book · **ANALYSIS:** = inference / standard knowledge · **CURRENT:** = current affairs. *Companion:* advanced/06_Himalayan-Glaciers-GLOF.md.

1. Snapshot & core idea

Foundation - Glacial Erosion & Deposition Landforms (GC Leong / Majid).

FACT: Moving ice erodes by plucking (freeze-thaw prising rock) and abrasion (scratching/grinding with embedded debris), then deposits its load as moraines and drift when it melts. A valley glacier fed by several corries flows downslope, gouges its floor and walls, and on melting drops unsorted till (moraine) and, via meltwater, sorted outwash.

- **FACT:** Cirque (corrie/cwm): plucking on the back-wall + abrasion of the floor deepen a horse-shoe basin; a small lake in it is a tarn.
- **FACT:** Arete = sharp ridge separating two cirques ('fish-bone'); where three or more cirques meet, a pyramidal horn forms (classic: Matterhorn).
- **FACT:** U-shaped valley (glacial trough): the glacier grinds a former V-valley into a broad, steep-sided U; tributary valleys are left as hanging valleys (waterfalls).
- **FACT:** Roche moutonnee: asymmetric bump - gently abraded up-ice side, plucked steep down-ice side - a hallmark of glacial erosion; a drowned trough is a fjord.
- **FACT:** Depositional (Majid): moraines (lateral, medial, terminal, ground) of unsorted till; drumlins (oval till mounds), eskers (meltwater ridges), and outwash plains/sandur of sorted sediment.

Ca Anchor - Gangotri Glacier Retreat (Gomukh Snout).

FACT: A retreating glacier snout is the visible edge of a shrinking cirque/valley-glacier system - directly linking glacial geomorphology to climate change and water security.

- **FACT:** The glacier terminus is the snout (here Gomukh), from which the Bhagirathi (headstream of the Ganga) emerges.
- **FACT:** Retreat exposes fresh moraine and pro-glacial ground; rates vary by period and measurement method, so quote a dated survey rather than one timeless annual rate.
- **FACT:** Hydrological impact is phased: meltwater may first increase, then decline after substantial ice loss; basin rainfall and seasonal snow remain crucial to Himalayan river flow.

2. Key classification / data

Type	Landform	What it is / diagnostic feature
FACT: Erosional	Cirque / corrie / cwm	Horse-shoe armchair basin at valley head; tarn may occupy it
FACT: Erosional	Arête	Sharp ridge between adjacent cirques/glacial valleys; French <i>arête</i> means ridge/edge
FACT: Erosional	Horn	Pyramidal peak where multiple cirques cut back, e.g. Matterhorn in source note
FACT: Erosional	U-shaped valley / glacial trough	Broad, steep-sided trough produced as ice abrades valley floor and sides
FACT: Erosional	Hanging valley	Tributary valley left above the main trough; waterfalls common
FACT: Erosional	Roche moutonnee	Asymmetric rock bump: abraded up-ice side, plucked steep down-ice side
FACT: Erosional	Fjord / fiord	Drowned, steep-sided glacial trough reaching the sea
FACT: Depositional	Moraines	Unsorted till: lateral, medial, terminal and ground moraines
FACT: Depositional	Drumlin	Oval mound of glacial drift/till formed beneath ice
FACT: Depositional	Esker / outwash plain	Sinuuous meltwater ridge / sorted sandur deposit

3. Study links

Study link: FACT: Geography -> Physical -> Glacial Landforms (GC Leong: Ice; Majid: Glacial Geomorphology)

Study link: FACT: Geography -> Glaciation -> Glacier Retreat (applied CA)

4. Must-Know Facts (Prelims)

Foundation - Glacial Erosion & Deposition Landforms (GC Leong / Majid).

- FACT: Glacial erosion = plucking + abrasion.
- FACT: Cirque -> tarn; arete = ridge; horn = pyramidal peak (Matterhorn).
- FACT: U-shaped valley + hanging valleys are glacial (rivers make V-valleys).
- FACT: Roche moutonnee: gentle up-ice, steep plucked down-ice.
- FACT: Moraine = unsorted till; outwash plain (sandur) = sorted meltwater deposit.
- FACT: Drumlin (till mound) vs esker (meltwater ridge).

Ca Anchor - Gangotri Glacier Retreat (Gomukh Snout).

- FACT: Gangotri snout = Gomukh; source of Bhagirathi -> Ganga.
- FACT: Long-term retreat is documented, but distance/rate must be tied to the survey period and method.
- FACT: Glacier retreat = shrinking valley/cirque glacier (geomorphology + climate).

5. UPSC Traps

MEMORY: Trap: Do not confuse similar-looking landforms, processes, scales or Indian locations; UPSC often tests the exact mechanism and setting.

Foundation - Glacial Erosion & Deposition Landforms (GC Leong / Majid).

- **WRONG:** U-shaped valleys are made by rivers. -> Rivers make V-shaped valleys; glaciers carve U-shaped troughs.
- **WRONG:** Moraine and outwash are both sorted. -> Moraine (till) is **UNSORTED**; outwash (meltwater) is **SORTED**.
- **WRONG:** A tarn is a depositional landform. -> A tarn is a lake in an **EROSIONAL** cirque basin.

Ca Anchor - Gangotri Glacier Retreat (Gomukh Snout).

- WRONG: The Ganga rises at Gangotri town. -> The Bhagirathi emerges from the GOMUKH snout of the Gangotri glacier.

6. CURRENT: Current link

Foundation - Glacial Erosion & Deposition Landforms (GC Leong / Majid). CURRENT: Glacial landforms explain the high Himalaya's scenery and its water towers - the same glaciers now retreating (Gangotri) and spawning hazardous glacial lakes.

Ca Anchor - Gangotri Glacier Retreat (Gomukh Snout). CURRENT: Gangotri's monitored snout retreat is a current cryosphere anchor. Treat the Bhagirathi as the headstream emerging at Gomukh and avoid converting glacier retreat directly into a simple, immediate "river will cease" claim.

7. Mains angles

Foundation - Glacial Erosion & Deposition Landforms (GC Leong / Majid).

- ANALYSIS: Distinguish glacial erosional vs depositional landforms and use them to read the physiography and water-tower role of the high Himalaya.

Ca Anchor - Gangotri Glacier Retreat (Gomukh Snout).

- ANALYSIS: Connect glacial geomorphology to climate-driven retreat and India's water-security stakes.

8. Glacier mass balance, ice masses and glacial-lake hazard

Why this section exists: the official syllabus asks for *changes in critical geographical features, including water bodies and ice caps, and the effects of such changes*. This file taught glacial landforms but not **why ice advances or retreats**, not the **types of ice mass**, and not the **hazard chain** that follows retreat. Those are the parts a 15-mark answer needs.

8.1 Mass balance: the accounting that decides advance and retreat

```
ACCUMULATION (snowfall, avalanche input, refreezing)  <-- gains, upper glacier
      |
      |
EQUILIBRIUM LINE (accumulation = ablation)
      |
      |
ABLATION (melting, sublimation, calving, wind scour)  <-- losses, lower glacier

NET BALANCE POSITIVE -> the equilibrium line falls -> the snout ADVANCES
NET BALANCE NEGATIVE -> the equilibrium line rises -> the snout RETREATS
```

- ANALYSIS: **A retreating glacier is still flowing forward.** Ice continues to move down-valley under gravity; the *snout* retreats because loss at the terminus outpaces delivery. This distinction is routinely mis-stated and is easy marks when written correctly.
- ANALYSIS: **Response is lagged and size-dependent:** small, steep, low-altitude glaciers respond within years; large, thick, high-altitude and debris-covered glaciers respond over decades. A single warm or snowy year therefore proves nothing about trend.
- ANALYSIS: **Debris cover complicates the signal:** a thin ash or dust layer absorbs heat and accelerates melt, while a thick debris mantle *insulates* the ice and slows it. Himalayan glaciers with heavy debris cover can therefore thin without the snout retreating visibly.

8.2 The family of ice masses

Ice mass	Form and setting	Diagnostic point
ANALYSIS: Valley (alpine) glacier	Confined by valley walls; flows down a pre-existing valley	Produces the U-shaped valley, cirque, arete and horn suite taught above
ANALYSIS: Cirque / corrie glacier	Small, occupying an armchair hollow	The nursery form; may be all that survives after retreat
ANALYSIS: Piedmont glacier	Valley glaciers spreading out on a lowland at the mountain front	Transitional between confined and unconfined flow
ANALYSIS: Ice cap	Dome of ice covering an upland, flowing outward radially, smaller in extent	Buries the relief rather than following it
ANALYSIS: Ice sheet	Continental-scale ice cap	Only Greenland and Antarctica today; the Pleistocene sheets created the landforms of northern Europe and North America
ANALYSIS: Ice shelf	Floating extension of an ice sheet over the sea	Already floating, so its collapse does not itself raise sea level - but it buttresses grounded ice behind it, and losing that buttress accelerates discharge that does
ANALYSIS: Iceberg	Calved floating fragment	Same floating-body logic as the shelf

MEMORY: **Trap**: floating ice melting does **not** directly raise sea level; **grounded** ice (glaciers, ice caps and the grounded parts of ice sheets) transferring mass to the ocean does. Sea-level mechanism is developed in 10_Coastal-Landforms.md.

8.3 Glacial lakes and the GLOF chain

GLACIER RETREAT

- > meltwater ponds behind the terminal/lateral MORaine or in the vacated basin
- > a PROGLACIAL or MORaine-DAMMED LAKE grows
- > the dam is unconsolidated debris, often with buried ice, and is inherently weak
- > a TRIGGER arrives: rock or ice avalanche into the lake, extreme rainfall, dam seepage and piping, earthquake, or upstream lake failure in cascade
- > SUDDEN BREACH -> GLACIAL LAKE OUTBURST FLOOD (GLOF)
- > a high-energy, sediment-charged flood wave travels far down a steep valley, scouring channels and destroying roads, bridges, settlements and hydropower assets

- ANALYSIS: **Why the Himalaya is exposed**: young, steep, seismically active relief; heavy debris supply; monsoon-concentrated rainfall; rapid downstream development of roads, towns and run-of-river hydropower in narrow valleys. The hazard has grown through **both** cryospheric change and increased exposure.

- ANALYSIS: **Cascading risk is the modern point**: a GLOF can be initiated by a mass movement rather than by melt alone, and the flood can trigger further failures downstream - so a single-cause framing ("warming caused the flood") is analytically weak.

- ANALYSIS: **Risk-reduction logic**: inventory and remote-sensing monitoring of lakes; controlled lowering or siphoning of high-risk lakes; early-warning and instrumented telemetry; hazard-zoned valley-floor land use and siting of infrastructure; and design standards for hydropower in glacier-fed catchments. Institutional doctrine belongs to Disaster -Management.

ANALYSIS: **Factual caution**: do **not** quote a Himalayan glacier count, a retreat rate in metres per year, an inventory of "potentially dangerous" lakes, a glacier-area-loss percentage, or the death toll or date of a specific disaster from memory. Each of these varies by inventory, method and period. Name the mechanism; attach a figure only with a cited source and its survey period.

8.4 Why ice matters far beyond the mountains

Effect of ice loss	Mechanism	Geographic consequence
ANALYSIS: Altered river regime	Reduced buffering of dry-season and drought-year flow	Himalayan rivers lose their late-season melt contribution; irrigation and hydropower reliability falls in exactly the driest months
ANALYSIS: Sea-level contribution	Grounded ice transfers mass to the ocean	Deltas, low islands and coastal cities - see 10_■Coastal-■Landforms.md and 11_■Islands-■and-■Coral-■Reefs.md
ANALYSIS: Albedo feedback	Bright ice replaced by darker rock or water absorbs more radiation	Local and regional warming amplification - see 25_■Arctic-■or-■Polar-■Climate.md
ANALYSIS: Slope destabilisation	Loss of ice support and permafrost thaw in rock walls	More rockfall and debris flow in deglaciating terrain
ANALYSIS: Sediment and hazard flux	Newly exposed moraine is easily mobilised	Increased sediment load, channel change and reservoir siltation downstream

9. Answer architecture (10/15/20-mark support)

9.1 Directive decoding for this topic

If the question says	It is really asking for	Do not
"Explain the retreat of Himalayan glaciers and its consequences"	Mass-balance logic, lag and debris-cover nuance, then the water-hazard-sea-level consequence chain	Assert an uncited retreat rate
"Discuss glacial lake outburst floods"	The dam-weakness-plus-trigger chain and the exposure half of the risk	Treat GLOF as ordinary flooding
"Distinguish glacial erosional and depositional landforms"	Process-based pairing, with the U-shaped valley versus V-shaped valley discriminator	List names without the process
"Changes in ice caps and their effects"	Ice-mass typology, the floating-versus-grounded rule, and the four effect pathways	Merge glaciers, ice shelves and sea ice into one category

9.2 Reusable 15-mark spine - "Himalayan glacier change and its effects"

1. **Thesis:** the significance of Himalayan glacier change lies less in the ice itself than in the **seasonal water-storage service** it performs for the most densely populated river basins on Earth - and in the hazard it releases while retreating.
2. **Mechanism:** mass balance and the equilibrium line; why the snout retreats while ice still flows; why response is lagged and heterogeneous across the range.
3. **Complication that shows command:** debris cover, elevation and size make a single range-wide statement unsafe; thinning and snout retreat are different measures.
4. **Effects, in descending order of certainty:** altered dry-season flow reliability; expanding glacial lakes and GLOF exposure; slope instability and sediment flux; contribution to sea-level rise via grounded ice.
5. **Balance:** monsoon rainfall, not melt, supplies the bulk of annual flow in most of these basins, so the melt contribution matters disproportionately in the **lean season and in drought years** rather than in the annual total. Saying this prevents the common exaggeration.
6. **Response:** monitoring and inventory, lake risk-reduction engineering, early warning, hazard-informed siting, and basin-scale storage and demand management.

7. **Conclusion:** graded - the water risk is a *timing and reliability* risk, and the hazard risk is a *joint product of cryospheric change and valley-floor development*.

9.3 Evidence units available in this file

Claim: landform evidence allows past climates to be reconstructed where no instrument record exists. **Evidence:** U-shaped valleys, cirques, aretes, horns and erratics occur far outside today's glaciated areas - including across northern Europe and North America. **Significance:** it proves that ice extent has changed radically, independent of any modern measurement. **Limitation:** landforms record *that* ice was present, not precisely when or for how long, without separate dating evidence.

Claim: a retreating glacier creates a hazard as well as losing a resource. **Evidence:** the moraine-dammed lakes left behind are impounded by unconsolidated debris, sometimes containing buried ice, and fail catastrophically when triggered. **Significance:** it links a slow climatic process to a sudden disaster, which is precisely the analytical link examiners reward. **Limitation:** the trigger is often a mass movement, so attributing any given outburst wholly to warming overstates the evidence.

Historical PYQ Integration (2018-2023)

Status: Question-level PYQ demand is integrated into this owner. **Provenance:** Audited local official-paper routing ledgers: _PYQ-ROUTING-MAINS-GS1-GS2-ESSAY-2018-2023.md, _PYQ-ROUTING-PRELIMS-2018-2023.md. **Answer-key rule:** The official 2018-2023 Prelims/CSAT keys are not held locally; no option or answer has been inferred.

- **Years represented:** 2019, 2023
- **Paper(s):** GS-I, Prelims GS-I
- **Routed question demands:** 2

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2019	Prelims GS-I	38	Himalayan rivers and associated glacier sources matching	Objective question; official key unavailable locally	Routed; key unavailable locally	Cover the named fact/concept and its likely statement-level distinctions.
2023	GS-I	6	Formation of fjords and their picturesque character	How are they formed and Why · 10 marks · 150 words	Routed to owning topic	Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion.

What this owner must now support

- Himalayan rivers and associated glacier sources matching
- Formation of fjords and their picturesque character

The table integrates the examinable demand and paper metadata. It does not turn an unkeyed objective question into a solved answer, and it does not claim that lexical presence alone proves full conceptual sufficiency.

Semantic-completeness ownership and PYQ control

- **Owned core:** glacier formation and movement; accumulation, ablation and mass balance; equilibrium-line altitude, snowline, firn line and terminus; plucking, abrasion, glacial erosion and deposition; cirque-arête-horn, U-shaped trough, hanging valley, roche moutonnée, fjord, till, moraines, drumlins, eskers and outwash; Himalayan glacier-river mapping, glacial lakes and GLOF risk.
- **Process control:** snowfall and refreezing -> firn and flowing ice -> plucking/abrasion and debris transport -> till or meltwater deposition. Glacier retreat or overdeepening can create/expand a lake; slope/ice entry, inflow, piping or erosion can produce overtopping or dam failure and a

sediment-laden downstream flood.

- **Scale/map control:** snow grain, glacier, catchment, Himalayan sector, glacial-lake dam and downstream valley are separate scales. Gangotri- Bhagirathi, Zemu-Teesta, Bara Shigri-Chenab and Siachen-Nubra-Shyok-Indus are map hooks, not claims about political boundaries or uniform basin response.
- **Date/data control:** glacier or lake counts, areas, retreat rates, mass balance, hazard classes and monitoring lists require inventory boundary, sensor/method, observation period and issuing agency. A dated inventory is not a present-day rate or proof that every mapped lake is dangerous.
- **Terminology control:** snowfield differs from glacier; ELA from climatic snowline, firn line and snout; retreat from reversal of ice flow; till from sorted outwash; fjord from ria; glacial-lake hazard from downstream risk.
- **Causal control:** warming can alter glacier and lake conditions but does not by itself establish an event-specific GLOF cause. Dam material, freeboard, slope instability, displacement waves, inflow, valley geometry, exposure, warning and infrastructure vulnerability must be tested.
- **Boundary:** Topic 05 owns river-system and basin-transfer analysis; Topic 09 owns general lake/wetland classification; Topic 10 owns relative sea-level and coasts; Disaster Management owns response doctrine. Topic 06 retains the cryosphere process, Himalayan map and GLOF hazard-to-risk chain.
- **Verified PYQ ownership, 2018-2026:** direct routes include the 2019 glacier-river matching objective demand, 2020 Himalayan glacier-loss and Indian water-resources Mains demand, and 2023 fjord Mains demand. No unavailable objective key, glacier count, retreat rate or hazard ranking is invented.

GEOGRAPHY DEEP-REVIEW CORE CONTROL

- **Must remember:** Glaciation proceeds through accumulation-ablation balance, ice deformation/basal motion, erosion, transport and deposition; cirque-arête-horn, U-shaped valley, moraine and outwash logic must connect to Himalayan mass balance, proglacial lakes and GLOF cascades.
- **Close distinction:** Snowline, equilibrium-line altitude, firn line and glacier terminus are distinct; till is unsorted ice-laid sediment while outwash is meltwater-sorted, and a fjord is a drowned glacial trough rather than any steep inlet.
- **Evidence / causal limit:** Retreat varies by glacier and period; lake growth alone does not prove imminent outburst, while dam material, freeboard, slope instability, ice/rock avalanches and downstream exposure condition risk.

BASIC MCQS / REMEDIATION

Q1. Which statement correctly explains Glacier definition?

A. A glacier is compacted and recrystallised land ice thick enough to deform or slide under gravity; a persistent snowfield need not be a glacier because it may not flow. B. The equilibrium-line altitude balances annual accumulation and ablation; the climatic snowline is a long-term regional limit, while the snout is the glacier terminus. C. Ice can continue moving down-valley while the snout retreats when ablation at the terminus exceeds forward ice delivery. D. Accumulation adds snow and refrozen water, ablation removes mass, and net balance is their difference over a stated glacier and period.

Answer: A. Explanation: A glacier is compacted and recrystallised land ice thick enough to deform or slide under gravity; a persistent snowfield need not be a glacier because it may not flow. The other options describe different processes, locations, scales or governance categories.

Q2. Which option is the safest spatial interpretation of Glacier definition?

A. Thin debris can enhance melt by lowering albedo, whereas sufficiently thick debris can insulate; debris-covered tongues may thin without rapid visible snout retreat. B. A glacier is compacted and recrystallised land ice thick enough to deform or slide under gravity; a persistent snowfield need not be a glacier because it may not flow. C. The equilibrium-line altitude balances annual accumulation and ablation; the climatic snowline is a long-term regional limit, while the snout is the glacier terminus. D. Ice can continue moving down-valley while the snout retreats when ablation at the terminus exceeds forward ice delivery.

Answer: B. Explanation: A glacier is compacted and recrystallised land ice thick enough to deform or slide under gravity; a persistent snowfield need not be a glacier because it may not flow. The other options describe different processes, locations, scales or governance categories.

Q3. Which statement preserves the process boundary for Glacier definition?

A. Ice can continue moving down-valley while the snout retreats when ablation at the terminus exceeds forward ice delivery. B. Plucking detaches jointed blocks and abrasion by entrained debris polishes, striates and grinds bedrock, producing rock flour and asymmetric roche moutonnee forms. C. A glacier is compacted and recrystallised land ice thick enough to deform or slide under gravity; a persistent snowfield need not be a glacier because it may not flow. D. Thin debris can enhance melt by lowering albedo, whereas sufficiently thick debris can insulate; debris-covered tongues may thin without rapid visible snout retreat.

Answer: C. Explanation: A glacier is compacted and recrystallised land ice thick enough to deform or slide under gravity; a persistent snowfield need not be a glacier because it may not flow. The other options describe different processes, locations, scales or governance categories.

Q4. Which option avoids the main UPSC trap concerning Glacier definition?

A. Plucking detaches jointed blocks and abrasion by entrained debris polishes, striates and grinds bedrock, producing rock flour and asymmetric roche moutonnee forms. B. Thin debris can enhance melt by lowering albedo, whereas sufficiently thick debris can insulate; debris-covered tongues may thin without rapid visible snout retreat. C. Cirques, aretes and horns develop around glacier heads; valley ice widens and deepens V-valleys into U-shaped troughs and leaves hanging tributary valleys. D. A glacier is compacted and recrystallised land ice thick enough to deform or slide under gravity; a persistent snowfield need not be a glacier because it may not flow.

Answer: D. Explanation: A glacier is compacted and recrystallised land ice thick enough to deform or slide under gravity; a persistent snowfield need not be a glacier because it may not flow. The other options describe different processes, locations, scales or governance categories.

Q5. Which statement correctly explains Mass-balance system?

A. Accumulation adds snow and refrozen water, ablation removes mass, and net balance is their difference over a stated glacier and period. B. Thin debris can enhance melt by lowering albedo, whereas sufficiently thick debris can insulate; debris-covered tongues may thin without rapid visible snout retreat. C. Ice can continue moving down-valley while the snout retreats when ablation at the terminus exceeds forward ice delivery. D. The equilibrium-line altitude balances annual accumulation and ablation; the climatic snowline is a long-term regional limit, while the snout is the glacier terminus.

Answer: A. Explanation: Accumulation adds snow and refrozen water, ablation removes mass, and net balance is their difference over a stated glacier and period. The other options describe different processes, locations, scales or governance categories.

Q6. Which option is the safest spatial interpretation of Mass-balance system?

A. Thin debris can enhance melt by lowering albedo, whereas sufficiently thick debris can insulate; debris-covered tongues may thin without rapid visible snout retreat. B. Accumulation adds snow and refrozen water, ablation removes mass, and net balance is their difference over a stated glacier and period. C. Plucking detaches jointed blocks and abrasion by entrained debris polishes, striates and grinds bedrock, producing rock flour and asymmetric roche moutonnée forms. D. Ice can continue moving down-valley while the snout retreats when ablation at the terminus exceeds forward ice delivery.

Answer: B. Explanation: Accumulation adds snow and refrozen water, ablation removes mass, and net balance is their difference over a stated glacier and period. The other options describe different processes, locations, scales or governance categories.

Q7. Which statement preserves the process boundary for Mass-balance system?

A. Plucking detaches jointed blocks and abrasion by entrained debris polishes, striates and grinds bedrock, producing rock flour and asymmetric roche moutonnée forms. B. Thin debris can enhance melt by lowering albedo, whereas sufficiently thick debris can insulate; debris-covered tongues may thin without rapid visible snout retreat. C. Accumulation adds snow and refrozen water, ablation removes mass, and net balance is their difference over a stated glacier and period. D. Cirques, aretes and horns develop around glacier heads; valley ice widens and deepens V-valleys into U-shaped troughs and leaves hanging tributary valleys.

Answer: C. Explanation: Accumulation adds snow and refrozen water, ablation removes mass, and net balance is their difference over a stated glacier and period. The other options describe different processes, locations, scales or governance categories.

Q8. Which option avoids the main UPSC trap concerning Mass-balance system?

A. A fjord is a marine-flooded glacial trough, whereas a ria is a drowned river valley; marine inundation follows rather than creates the glacial excavation. B. Plucking detaches jointed blocks and abrasion by entrained debris polishes, striates and grinds bedrock, producing rock flour and asymmetric roche moutonnée forms. C. Cirques, aretes and horns develop around glacier heads; valley ice widens and deepens V-valleys into U-shaped troughs and leaves hanging tributary valleys. D. Accumulation adds snow and refrozen water, ablation removes mass, and net balance is their difference over a stated glacier and period.

Answer: D. Explanation: Accumulation adds snow and refrozen water, ablation removes mass, and net balance is their difference over a stated glacier and period. The other options describe different processes, locations, scales or governance categories.

Q9. Which statement correctly explains ELA, snowline and snout?

A. The equilibrium-line altitude balances annual accumulation and ablation; the climatic snowline is a long-term regional limit, while the snout is the glacier terminus. B. Plucking detaches jointed blocks and abrasion by entrained debris polishes, striates and grinds bedrock, producing rock flour and asymmetric roche moutonnée forms. C. Thin debris can enhance melt by lowering albedo, whereas sufficiently thick debris can insulate; debris-covered tongues may thin without rapid visible snout retreat. D. Ice can continue moving down-valley while the snout retreats when ablation at the terminus exceeds forward ice delivery.

Answer: A. Explanation: The equilibrium-line altitude balances annual accumulation and ablation; the climatic snowline is a long-term regional limit, while the snout is the glacier terminus. The other options describe different processes, locations, scales or governance categories.

Q10. Which option is the safest spatial interpretation of ELA, snowline and snout?

A. Thin debris can enhance melt by lowering albedo, whereas sufficiently thick debris can insulate; debris-covered tongues may thin without rapid visible snout retreat. B. The equilibrium-line altitude balances annual accumulation and ablation; the climatic snowline is a long-term regional limit, while the snout is the glacier terminus. C. Cirques, aretes and horns develop around glacier heads; valley ice widens and deepens V-valleys into U-shaped troughs and leaves hanging tributary valleys. D. Plucking detaches jointed blocks and abrasion by entrained debris polishes, striates and grinds bedrock, producing rock flour and asymmetric roche moutonnée forms.

Answer: B. Explanation: The equilibrium-line altitude balances annual accumulation and ablation; the climatic snowline is a long-term regional limit, while the snout is the glacier terminus. The other options describe different processes, locations, scales or governance categories.

Q11. Which statement preserves the process boundary for ELA, snowline and snout?

A. A fjord is a marine-flooded glacial trough, whereas a ria is a drowned river valley; marine inundation follows rather than creates the glacial excavation. B. Plucking detaches jointed blocks and abrasion by entrained debris polishes, striates and grinds bedrock, producing rock flour and asymmetric roche moutonnée forms. C. The equilibrium-line altitude balances annual accumulation and ablation; the climatic snowline is a long-term regional limit, while the snout is the glacier terminus. D. Cirques, aretes and horns develop around glacier heads; valley ice widens and deepens V-valleys into U-shaped troughs and leaves hanging tributary valleys.

Answer: C. Explanation: The equilibrium-line altitude balances annual accumulation and ablation; the climatic snowline is a long-term regional limit, while the snout is the glacier terminus. The other options describe different processes, locations, scales or governance categories.

Q12. Which option avoids the main UPSC trap concerning ELA, snowline and snout?

A. Cirques, aretes and horns develop around glacier heads; valley ice widens and deepens V-valleys into U-shaped troughs and leaves hanging tributary valleys. B. A fjord is a marine-flooded glacial trough, whereas a ria is a drowned river valley; marine inundation follows rather than creates the glacial excavation. C. Direct ice deposition is generally unsorted till; lateral, medial, terminal and ground moraines describe debris position or form. D. The equilibrium-line altitude balances annual accumulation and ablation; the climatic snowline is a long-term regional limit, while the snout is the glacier terminus.

Answer: D. Explanation: The equilibrium-line altitude balances annual accumulation and ablation; the climatic snowline is a long-term regional limit, while the snout is the glacier terminus. The other options describe different processes, locations, scales or governance categories.

Q13. Which statement correctly explains Forward-flow retreat paradox?

A. Ice can continue moving down-valley while the snout retreats when ablation at the terminus exceeds forward ice delivery. B. Thin debris can enhance melt by lowering albedo, whereas sufficiently thick debris can insulate; debris-covered tongues may thin without rapid visible snout retreat. C. Cirques, aretes and horns develop around glacier heads; valley ice widens and deepens V-valleys into U-shaped troughs and leaves hanging tributary valleys. D. Plucking detaches jointed blocks and abrasion by entrained debris polishes, striates and grinds bedrock, producing rock flour and asymmetric roche moutonnée forms.

Answer: A. Explanation: Ice can continue moving down-valley while the snout retreats when ablation at the terminus exceeds forward ice delivery. The other options describe different processes, locations, scales or governance categories.

Q14. Which option is the safest spatial interpretation of Forward-flow retreat paradox?

A. Cirques, aretes and horns develop around glacier heads; valley ice widens and deepens V-valleys into U-shaped troughs and leaves hanging tributary valleys. B. Ice can continue moving down-valley while the snout retreats when ablation at the terminus exceeds forward ice delivery. C. Plucking detaches jointed blocks and abrasion by entrained debris polishes, striates and grinds bedrock, producing rock flour and asymmetric roche moutonnee forms. D. A fjord is a marine-flooded glacial trough, whereas a ria is a drowned river valley; marine inundation follows rather than creates the glacial excavation.

Answer: B. Explanation: Ice can continue moving down-valley while the snout retreats when ablation at the terminus exceeds forward ice delivery. The other options describe different processes, locations, scales or governance categories.

Q15. Which statement preserves the process boundary for Forward-flow retreat paradox?

A. A fjord is a marine-flooded glacial trough, whereas a ria is a drowned river valley; marine inundation follows rather than creates the glacial excavation. B. Direct ice deposition is generally unsorted till; lateral, medial, terminal and ground moraines describe debris position or form. C. Ice can continue moving down-valley while the snout retreats when ablation at the terminus exceeds forward ice delivery. D. Cirques, aretes and horns develop around glacier heads; valley ice widens and deepens V-valleys into U-shaped troughs and leaves hanging tributary valleys.

Answer: C. Explanation: Ice can continue moving down-valley while the snout retreats when ablation at the terminus exceeds forward ice delivery. The other options describe different processes, locations, scales or governance categories.

Q16. Which option avoids the main UPSC trap concerning Forward-flow retreat paradox?

A. A fjord is a marine-flooded glacial trough, whereas a ria is a drowned river valley; marine inundation follows rather than creates the glacial excavation. B. Eskers are sinuous ridges deposited by subglacial or englacial streams, while outwash plains contain relatively sorted sediment spread by braided meltwater. C. Direct ice deposition is generally unsorted till; lateral, medial, terminal and ground moraines describe debris position or form. D. Ice can continue moving down-valley while the snout retreats when ablation at the terminus exceeds forward ice delivery.

Answer: D. Explanation: Ice can continue moving down-valley while the snout retreats when ablation at the terminus exceeds forward ice delivery. The other options describe different processes, locations, scales or governance categories.

Q17. Which statement correctly explains Debris-cover qualification?

A. Thin debris can enhance melt by lowering albedo, whereas sufficiently thick debris can insulate; debris-covered tongues may thin without rapid visible snout retreat. B. Cirques, aretes and horns develop around glacier heads; valley ice widens and deepens V-valleys into U-shaped troughs and leaves hanging tributary valleys. C. Plucking detaches jointed blocks and abrasion by entrained debris polishes, striates and grinds bedrock, producing rock flour and asymmetric roche moutonnee forms. D. A fjord is a marine-flooded glacial trough, whereas a ria is a drowned river valley; marine inundation follows rather than creates the glacial excavation.

Answer: A. Explanation: Thin debris can enhance melt by lowering albedo, whereas sufficiently thick debris can insulate; debris-covered tongues may thin without rapid visible snout retreat. The other options describe different processes, locations, scales or governance categories.

Q18. Which option is the safest spatial interpretation of Debris-cover qualification?

A. Cirques, aretes and horns develop around glacier heads; valley ice widens and deepens V-valleys into U-shaped troughs and leaves hanging tributary valleys. B. Thin debris can enhance melt by lowering albedo, whereas sufficiently thick debris can insulate; debris-covered tongues may thin without rapid visible snout retreat. C. A fjord is a marine-flooded glacial trough, whereas a ria is a drowned river valley; marine inundation follows rather than creates the glacial excavation. D. Direct ice deposition is generally unsorted till; lateral, medial, terminal and ground moraines describe debris position or form.

Answer: B. Explanation: Thin debris can enhance melt by lowering albedo, whereas sufficiently thick debris can insulate; debris-covered tongues may thin without rapid visible snout retreat. The other options describe

different processes, locations, scales or governance categories.

Q19. Which statement preserves the process boundary for Debris-cover qualification?

A. Direct ice deposition is generally unsorted till; lateral, medial, terminal and ground moraines describe debris position or form. B. Eskers are sinuous ridges deposited by subglacial or englacial streams, while outwash plains contain relatively sorted sediment spread by braided meltwater. C. Thin debris can enhance melt by lowering albedo, whereas sufficiently thick debris can insulate; debris-covered tongues may thin without rapid visible snout retreat. D. A fjord is a marine-flooded glacial trough, whereas a ria is a drowned river valley; marine inundation follows rather than creates the glacial excavation.

Answer: C. Explanation: Thin debris can enhance melt by lowering albedo, whereas sufficiently thick debris can insulate; debris-covered tongues may thin without rapid visible snout retreat. The other options describe different processes, locations, scales or governance categories.

Q20. Which option avoids the main UPSC trap concerning Debris-cover qualification?

A. Direct ice deposition is generally unsorted till; lateral, medial, terminal and ground moraines describe debris position or form. B. Loss of grounded glaciers, ice caps and ice sheets adds ocean mass; floating sea ice or shelf melt does not directly add mass beyond displacement, though shelves can buttress grounded ice. C. Eskers are sinuous ridges deposited by subglacial or englacial streams, while outwash plains contain relatively sorted sediment spread by braided meltwater. D. Thin debris can enhance melt by lowering albedo, whereas sufficiently thick debris can insulate; debris-covered tongues may thin without rapid visible snout retreat.

Answer: D. Explanation: Thin debris can enhance melt by lowering albedo, whereas sufficiently thick debris can insulate; debris-covered tongues may thin without rapid visible snout retreat. The other options describe different processes, locations, scales or governance categories.

Q21. Which statement correctly explains Glacial erosion?

A. Plucking detaches jointed blocks and abrasion by entrained debris polishes, striates and grinds bedrock, producing rock flour and asymmetric roche moutonnee forms. B. A fjord is a marine-flooded glacial trough, whereas a ria is a drowned river valley; marine inundation follows rather than creates the glacial excavation. C. Direct ice deposition is generally unsorted till; lateral, medial, terminal and ground moraines describe debris position or form. D. Cirques, aretes and horns develop around glacier heads; valley ice widens and deepens V-valleys into U-shaped troughs and leaves hanging tributary valleys.

Answer: A. Explanation: Plucking detaches jointed blocks and abrasion by entrained debris polishes, striates and grinds bedrock, producing rock flour and asymmetric roche moutonnee forms. The other options describe different processes, locations, scales or governance categories.

Q22. Which option is the safest spatial interpretation of Glacial erosion?

A. Eskers are sinuous ridges deposited by subglacial or englacial streams, while outwash plains contain relatively sorted sediment spread by braided meltwater. B. Plucking detaches jointed blocks and abrasion by entrained debris polishes, striates and grinds bedrock, producing rock flour and asymmetric roche moutonnee forms. C. Direct ice deposition is generally unsorted till; lateral, medial, terminal and ground moraines describe debris position or form. D. A fjord is a marine-flooded glacial trough, whereas a ria is a drowned river valley; marine inundation follows rather than creates the glacial excavation.

Answer: B. Explanation: Plucking detaches jointed blocks and abrasion by entrained debris polishes, striates and grinds bedrock, producing rock flour and asymmetric roche moutonnee forms. The other options describe different processes, locations, scales or governance categories.

Q23. Which statement preserves the process boundary for Glacial erosion?

A. Direct ice deposition is generally unsorted till; lateral, medial, terminal and ground moraines describe debris position or form. B. Eskers are sinuous ridges deposited by subglacial or englacial streams, while outwash plains contain relatively sorted sediment spread by braided meltwater. C. Plucking detaches jointed blocks and abrasion by entrained debris polishes, striates and grinds bedrock, producing rock flour and asymmetric roche moutonnee forms. D. Loss of grounded glaciers, ice caps and ice sheets adds ocean mass; floating sea ice or shelf melt does not directly add mass beyond displacement, though shelves can buttress grounded ice.

Answer: C. Explanation: Plucking detaches jointed blocks and abrasion by entrained debris polishes, striates and grinds bedrock, producing rock flour and asymmetric roche moutonnee forms. The other options describe different processes, locations, scales or governance categories.

Q24. Which option avoids the main UPSC trap concerning Glacial erosion?

A. Gangotri links to the Bhagirathi, Zemu to the Teesta, Bara Shigri to the Chenab system and Siachen to the Nubra-Shyok-Indus system; map claims must remain distinct from boundary claims. B. Loss of grounded glaciers, ice caps and ice sheets adds ocean mass; floating sea ice or shelf melt does not directly add mass beyond displacement, though shelves can buttress grounded ice. C. Eskers are sinuous ridges deposited by subglacial or englacial streams, while outwash plains contain relatively sorted sediment spread by braided meltwater. D. Plucking detaches jointed blocks and abrasion by entrained debris polishes, striates and grinds bedrock, producing rock flour and asymmetric roche moutonnee forms.

Answer: D. Explanation: Plucking detaches jointed blocks and abrasion by entrained debris polishes, striates and grinds bedrock, producing rock flour and asymmetric roche moutonnee forms. The other options describe different processes, locations, scales or governance categories.

Q25. Which statement correctly explains Erosional landform suite?

A. Cirques, aretes and horns develop around glacier heads; valley ice widens and deepens V-valleys into U-shaped troughs and leaves hanging tributary valleys. B. A fjord is a marine-flooded glacial trough, whereas a ria is a drowned river valley; marine inundation follows rather than creates the glacial excavation. C. Direct ice deposition is generally unsorted till; lateral, medial, terminal and ground moraines describe debris position or form. D. Eskers are sinuous ridges deposited by subglacial or englacial streams, while outwash plains contain relatively sorted sediment spread by braided meltwater.

Answer: A. Explanation: Cirques, aretes and horns develop around glacier heads; valley ice widens and deepens V-valleys into U-shaped troughs and leaves hanging tributary valleys. The other options describe different processes, locations, scales or governance categories.

Q26. Which option is the safest spatial interpretation of Erosional landform suite?

A. Loss of grounded glaciers, ice caps and ice sheets adds ocean mass; floating sea ice or shelf melt does not directly add mass beyond displacement, though shelves can buttress grounded ice. B. Cirques, aretes and horns develop around glacier heads; valley ice widens and deepens V-valleys into U-shaped troughs and leaves hanging tributary valleys. C. Direct ice deposition is generally unsorted till; lateral, medial, terminal and ground moraines describe debris position or form. D. Eskers are sinuous ridges deposited by subglacial or englacial streams, while outwash plains contain relatively sorted sediment spread by braided meltwater.

Answer: B. Explanation: Cirques, aretes and horns develop around glacier heads; valley ice widens and deepens V-valleys into U-shaped troughs and leaves hanging tributary valleys. The other options describe different processes, locations, scales or governance categories.

Q27. Which statement preserves the process boundary for Erosional landform suite?

A. Loss of grounded glaciers, ice caps and ice sheets adds ocean mass; floating sea ice or shelf melt does not directly add mass beyond displacement, though shelves can buttress grounded ice. B. Gangotri links to the Bhagirathi, Zemu to the Teesta, Bara Shigri to the Chenab system and Siachen to the Nubra-Shyok-Indus system; map claims must remain distinct from boundary claims. C. Cirques, aretes and horns develop around glacier heads; valley ice widens and deepens V-valleys into U-shaped troughs and leaves hanging tributary valleys. D. Eskers are sinuous ridges deposited by subglacial or englacial streams, while outwash plains contain relatively sorted sediment spread by braided meltwater.

Answer: C. Explanation: Cirques, aretes and horns develop around glacier heads; valley ice widens and deepens V-valleys into U-shaped troughs and leaves hanging tributary valleys. The other options describe different processes, locations, scales or governance categories.

Q28. Which option avoids the main UPSC trap concerning Erosional landform suite?

A. Gangotri links to the Bhagirathi, Zemu to the Teesta, Bara Shigri to the Chenab system and Siachen to the Nubra-Shyok-Indus system; map claims must remain distinct from boundary claims. B. Loss of grounded glaciers, ice caps and ice sheets adds ocean mass; floating sea ice or shelf melt does not directly add mass beyond displacement, though shelves can buttress grounded ice. C. Sustained ice loss can initially raise local melt contribution but later weaken lean-season and drought-year buffering; monsoon rain, snow and groundwater prevent a uniform all-rivers-dry conclusion. D. Cirques, aretes and horns develop around glacier heads; valley ice widens and deepens V-valleys into U-shaped troughs and leaves hanging tributary valleys.

Answer: D. Explanation: Cirques, aretes and horns develop around glacier heads; valley ice widens and deepens V-valleys into U-shaped troughs and leaves hanging tributary valleys. The other options describe different processes, locations, scales or governance categories.

Q29. Which statement correctly explains Fjord discriminator?

A. A fjord is a marine-flooded glacial trough, whereas a ria is a drowned river valley; marine inundation follows rather than creates the glacial excavation. B. Eskers are sinuous ridges deposited by subglacial or englacial streams, while outwash plains contain relatively sorted sediment spread by braided meltwater. C. Loss of grounded glaciers, ice caps and ice sheets adds ocean mass; floating sea ice or shelf melt does not directly add mass beyond displacement, though shelves can buttress grounded ice. D. Direct ice deposition is generally unsorted till; lateral, medial, terminal and ground moraines describe debris position or form.

Answer: A. Explanation: A fjord is a marine-flooded glacial trough, whereas a ria is a drowned river valley; marine inundation follows rather than creates the glacial excavation. The other options describe different processes, locations, scales or governance categories.

Q30. Which option is the safest spatial interpretation of Fjord discriminator?

A. Loss of grounded glaciers, ice caps and ice sheets adds ocean mass; floating sea ice or shelf melt does not directly add mass beyond displacement, though shelves can buttress grounded ice. B. A fjord is a marine-flooded glacial trough, whereas a ria is a drowned river valley; marine inundation follows rather than creates the glacial excavation. C. Eskers are sinuous ridges deposited by subglacial or englacial streams, while outwash plains contain relatively sorted sediment spread by braided meltwater. D. Gangotri links to the Bhagirathi, Zemu to the Teesta, Bara Shigri to the Chenab system and Siachen to the Nubra-Shyok-Indus system; map claims must remain distinct from boundary claims.

Answer: B. Explanation: A fjord is a marine-flooded glacial trough, whereas a ria is a drowned river valley; marine inundation follows rather than creates the glacial excavation. The other options describe different processes, locations, scales or governance categories.

Q31. Which statement preserves the process boundary for Fjord discriminator?

A. Loss of grounded glaciers, ice caps and ice sheets adds ocean mass; floating sea ice or shelf melt does not directly add mass beyond displacement, though shelves can buttress grounded ice. B. Sustained ice loss can initially raise local melt contribution but later weaken lean-season and drought-year buffering; monsoon rain, snow and groundwater prevent a uniform all-rivers-dry conclusion. C. A fjord is a marine-flooded glacial trough, whereas a ria is a drowned river valley; marine inundation follows rather than creates the glacial excavation. D. Gangotri links to the Bhagirathi, Zemu to the Teesta, Bara Shigri to the Chenab system and Siachen to the Nubra-Shyok-Indus system; map claims must remain distinct from boundary claims.

Answer: C. Explanation: A fjord is a marine-flooded glacial trough, whereas a ria is a drowned river valley; marine inundation follows rather than creates the glacial excavation. The other options describe different processes, locations, scales or governance categories.

Q32. Which option avoids the main UPSC trap concerning Fjord discriminator?

A. Gangotri links to the Bhagirathi, Zemu to the Teesta, Bara Shigri to the Chenab system and Siachen to the Nubra-Shyok-Indus system; map claims must remain distinct from boundary claims. B. Retreat can expose overdeepened basins or leave moraine dams that store water; lake existence alone does not establish instability or downstream risk. C. Sustained ice loss can initially raise local melt contribution but later weaken lean-season and drought-year buffering; monsoon rain, snow and groundwater prevent a uniform all-rivers-dry conclusion. D. A fjord is a marine-flooded glacial trough, whereas a ria is a drowned river valley; marine inundation follows rather than creates the glacial excavation.

Answer: D. Explanation: A fjord is a marine-flooded glacial trough, whereas a ria is a drowned river valley; marine inundation follows rather than creates the glacial excavation. The other options describe different processes, locations, scales or governance categories.

Q33. Which statement correctly explains Till and moraine?

A. Direct ice deposition is generally unsorted till; lateral, medial, terminal and ground moraines describe debris position or form. B. Loss of grounded glaciers, ice caps and ice sheets adds ocean mass; floating sea ice or shelf melt does not directly add mass beyond displacement, though shelves can buttress grounded ice. C. Gangotri links to the Bhagirathi, Zemu to the Teesta, Bara Shigri to the Chenab system and Siachen to the Nubra-Shyok-Indus system; map claims must remain distinct from boundary claims. D. Eskers are sinuous ridges deposited by subglacial or englacial streams, while outwash plains contain relatively sorted sediment spread by braided meltwater.

Answer: A. Explanation: Direct ice deposition is generally unsorted till; lateral, medial, terminal and ground moraines describe debris position or form. The other options describe different processes, locations, scales or governance categories.

Q34. Which option is the safest spatial interpretation of Till and moraine?

A. Sustained ice loss can initially raise local melt contribution but later weaken lean-season and drought-year buffering; monsoon rain, snow and groundwater prevent a uniform all-rivers-dry conclusion. B. Direct ice deposition is generally unsorted till; lateral, medial, terminal and ground moraines describe debris position or form. C. Gangotri links to the Bhagirathi, Zemu to the Teesta, Bara Shigri to the Chenab system and Siachen to the Nubra-Shyok-Indus system; map claims must remain distinct from boundary claims. D. Loss of grounded glaciers, ice caps and ice sheets adds ocean mass; floating sea ice or shelf melt does not directly add mass beyond displacement, though shelves can buttress grounded ice.

Answer: B. Explanation: Direct ice deposition is generally unsorted till; lateral, medial, terminal and ground moraines describe debris position or form. The other options describe different processes, locations, scales or governance categories.

Q35. Which statement preserves the process boundary for Till and moraine?

A. Retreat can expose overdeepened basins or leave moraine dams that store water; lake existence alone does not establish instability or downstream risk. B. Sustained ice loss can initially raise local melt contribution but later weaken lean-season and drought-year buffering; monsoon rain, snow and groundwater prevent a uniform all-rivers-dry conclusion. C. Direct ice deposition is generally unsorted till; lateral, medial, terminal and ground moraines describe debris position or form. D. Gangotri links to the Bhagirathi, Zemu to the Teesta, Bara Shigri to the Chenab system and Siachen to the Nubra-Shyok-Indus system; map claims must remain distinct from boundary claims.

Answer: C. Explanation: Direct ice deposition is generally unsorted till; lateral, medial, terminal and ground moraines describe debris position or form. The other options describe different processes, locations, scales or governance categories.

Q36. Which option avoids the main UPSC trap concerning Till and moraine?

A. Sustained ice loss can initially raise local melt contribution but later weaken lean-season and drought-year buffering; monsoon rain, snow and groundwater prevent a uniform all-rivers-dry conclusion. B. A GLOF is sudden lake release after overtopping or dam failure; triggers can include slope failure, ice or rock entry, intense inflow, piping, erosion or seismic disturbance. C. Retreat can expose overdeepened basins or leave moraine dams that store water; lake existence alone does not establish instability or downstream risk. D. Direct ice deposition is generally unsorted till; lateral, medial, terminal and ground moraines describe debris position or form.

Answer: D. Explanation: Direct ice deposition is generally unsorted till; lateral, medial, terminal and ground moraines describe debris position or form. The other options describe different processes, locations, scales or governance categories.

Q37. Which statement correctly explains Meltwater deposits?

A. Eskers are sinuous ridges deposited by subglacial or englacial streams, while outwash plains contain relatively sorted sediment spread by braided meltwater. B. Loss of grounded glaciers, ice caps and ice sheets adds ocean mass; floating sea ice or shelf melt does not directly add mass beyond displacement, though shelves can buttress grounded ice. C. Gangotri links to the Bhagirathi, Zemu to the Teesta, Bara Shigri to the Chenab system and Siachen to the Nubra-Shyok-Indus system; map claims must remain distinct from boundary claims. D. Sustained ice loss can initially raise local melt contribution but later weaken lean-season and drought-year buffering; monsoon rain, snow and groundwater prevent a uniform all-rivers-dry conclusion.

Answer: A. Explanation: Eskers are sinuous ridges deposited by subglacial or englacial streams, while outwash plains contain relatively sorted sediment spread by braided meltwater. The other options describe different processes, locations, scales or governance categories.

Q38. Which option is the safest spatial interpretation of Meltwater deposits?

A. Gangotri links to the Bhagirathi, Zemu to the Teesta, Bara Shigri to the Chenab system and Siachen to the Nubra-Shyok-Indus system; map claims must remain distinct from boundary claims. B. Eskers are sinuous ridges deposited by subglacial or englacial streams, while outwash plains contain relatively sorted sediment spread by braided meltwater. C. Sustained ice loss can initially raise local melt contribution but later weaken lean-season and drought-year buffering; monsoon rain, snow and groundwater prevent a uniform all-rivers-dry conclusion. D. Retreat can expose overdeepened basins or leave moraine dams that store water; lake existence alone does not establish instability or downstream risk.

Answer: B. Explanation: Eskers are sinuous ridges deposited by subglacial or englacial streams, while outwash plains contain relatively sorted sediment spread by braided meltwater. The other options describe different processes, locations, scales or governance categories.

Q39. Which statement preserves the process boundary for Meltwater deposits?

A. Sustained ice loss can initially raise local melt contribution but later weaken lean-season and drought-year buffering; monsoon rain, snow and groundwater prevent a uniform all-rivers-dry conclusion. B. A GLOF is sudden lake release after overtopping or dam failure; triggers can include slope failure, ice or rock entry, intense inflow, piping, erosion or seismic disturbance. C. Eskers are sinuous ridges deposited by subglacial or englacial streams, while outwash plains contain relatively sorted sediment spread by braided meltwater. D. Retreat can expose overdeepened basins or leave moraine dams that store water; lake existence alone does not establish instability or downstream risk.

Answer: C. Explanation: Eskers are sinuous ridges deposited by subglacial or englacial streams, while outwash plains contain relatively sorted sediment spread by braided meltwater. The other options describe different processes, locations, scales or governance categories.

Q40. Which option avoids the main UPSC trap concerning Meltwater deposits?

A. Retreat can expose overdeepened basins or leave moraine dams that store water; lake existence alone does not establish instability or downstream risk. B. A GLOF is sudden lake release after overtopping or dam failure; triggers can include slope failure, ice or rock entry, intense inflow, piping, erosion or seismic disturbance. C. GLOF hazard depends on lake, dam, slope and breach processes; risk additionally requires exposed people, settlements, roads, hydropower and limited warning or evacuation capacity. D. Eskers are sinuous ridges deposited by subglacial or englacial streams, while outwash plains contain relatively sorted sediment spread by braided meltwater.

Answer: D. Explanation: Eskers are sinuous ridges deposited by subglacial or englacial streams, while outwash plains contain relatively sorted sediment spread by braided meltwater. The other options describe different processes, locations, scales or governance categories.

Q41. Which statement correctly explains Ice-mass sea-level rule?

A. Loss of grounded glaciers, ice caps and ice sheets adds ocean mass; floating sea ice or shelf melt does not directly add mass beyond displacement, though shelves can buttress grounded ice. B. Retreat can expose overdeepened basins or leave moraine dams that store water; lake existence alone does not establish instability or downstream risk. C. Sustained ice loss can initially raise local melt contribution but later weaken lean-season and drought-year buffering; monsoon rain, snow and groundwater prevent a uniform all-rivers-dry conclusion. D. Gangotri links to the Bhagirathi, Zemu to the Teesta, Bara Shigri to the Chenab system and Siachen to the Nubra-Shyok-Indus system; map claims must remain distinct from boundary claims.

Answer: A. Explanation: Loss of grounded glaciers, ice caps and ice sheets adds ocean mass; floating sea ice or shelf melt does not directly add mass beyond displacement, though shelves can buttress grounded ice. The other options describe different processes, locations, scales or governance categories.

Q42. Which option is the safest spatial interpretation of Ice-mass sea-level rule?

A. Sustained ice loss can initially raise local melt contribution but later weaken lean-season and drought-year buffering; monsoon rain, snow and groundwater prevent a uniform all-rivers-dry conclusion. B. Loss of grounded glaciers, ice caps and ice sheets adds ocean mass; floating sea ice or shelf melt does not directly add mass beyond displacement, though shelves can buttress grounded ice. C. Retreat can expose overdeepened basins or leave moraine dams that store water; lake existence alone does not establish instability or downstream risk. D. A GLOF is sudden lake release after overtopping or dam failure; triggers can include slope failure, ice or rock entry, intense inflow, piping, erosion or seismic disturbance.

Answer: B. Explanation: Loss of grounded glaciers, ice caps and ice sheets adds ocean mass; floating sea ice or shelf melt does not directly add mass beyond displacement, though shelves can buttress grounded ice. The other options describe different processes, locations, scales or governance categories.

Q43. Which statement preserves the process boundary for Ice-mass sea-level rule?

A. A GLOF is sudden lake release after overtopping or dam failure; triggers can include slope failure, ice or rock entry, intense inflow, piping, erosion or seismic disturbance. B. GLOF hazard depends on lake, dam, slope and breach processes; risk additionally requires exposed people, settlements, roads, hydropower and limited warning or evacuation capacity. C. Loss of grounded glaciers, ice caps and ice sheets adds ocean mass; floating sea ice or shelf melt does not directly add mass beyond displacement, though shelves can buttress grounded ice. D. Retreat can expose overdeepened basins or leave moraine dams that store water; lake existence alone does not establish instability or downstream risk.

Answer: C. Explanation: Loss of grounded glaciers, ice caps and ice sheets adds ocean mass; floating sea ice or shelf melt does not directly add mass beyond displacement, though shelves can buttress grounded ice. The other options describe different processes, locations, scales or governance categories.

Q44. Which option avoids the main UPSC trap concerning Ice-mass sea-level rule?

A. A GLOF is sudden lake release after overtopping or dam failure; triggers can include slope failure, ice or rock entry, intense inflow, piping, erosion or seismic disturbance. B. South Lhonak was a moraine-dammed lake involved in the October 2023 Teesta disaster; event-specific work examines slope movement, impulse, overtopping and breach without reducing attribution to warming alone. C. GLOF hazard depends on lake, dam, slope and breach processes; risk additionally requires exposed people, settlements, roads, hydropower and limited warning or evacuation capacity. D. Loss of grounded glaciers, ice caps and ice sheets adds ocean mass; floating sea ice or shelf melt does not directly add mass beyond displacement, though shelves can buttress grounded ice.

Answer: D. Explanation: Loss of grounded glaciers, ice caps and ice sheets adds ocean mass; floating sea ice or shelf melt does not directly add mass beyond displacement, though shelves can buttress grounded ice. The other options describe different processes, locations, scales or governance categories.

Q45. Which statement correctly explains Himalayan map discipline?

A. Gangotri links to the Bhagirathi, Zemu to the Teesta, Bara Shigri to the Chenab system and Siachen to the Nubra-Shyok-Indus system; map claims must remain distinct from boundary claims. B. A GLOF is sudden lake release after overtopping or dam failure; triggers can include slope failure, ice or rock entry, intense inflow, piping, erosion or seismic disturbance. C. Retreat can expose overdeepened basins or leave moraine dams that store water; lake existence alone does not establish instability or downstream risk. D. Sustained ice loss can initially raise local melt contribution but later weaken lean-season and drought-year buffering; monsoon rain, snow and groundwater prevent a uniform all-rivers-dry conclusion.

Answer: A. Explanation: Gangotri links to the Bhagirathi, Zemu to the Teesta, Bara Shigri to the Chenab system and Siachen to the Nubra-Shyok-Indus system; map claims must remain distinct from boundary claims. The other options describe different processes, locations, scales or governance categories.

Q46. Which option is the safest spatial interpretation of Himalayan map discipline?

A. A GLOF is sudden lake release after overtopping or dam failure; triggers can include slope failure, ice or rock entry, intense inflow, piping, erosion or seismic disturbance. B. Gangotri links to the Bhagirathi, Zemu to the Teesta, Bara Shigri to the Chenab system and Siachen to the Nubra-Shyok-Indus system; map claims must remain distinct from boundary claims. C. Retreat can expose overdeepened basins or leave moraine dams that store water; lake existence alone does not establish instability or downstream risk. D. GLOF hazard depends on lake, dam, slope and breach processes; risk additionally requires exposed people, settlements, roads, hydropower and limited warning or evacuation capacity.

Answer: B. Explanation: Gangotri links to the Bhagirathi, Zemu to the Teesta, Bara Shigri to the Chenab system and Siachen to the Nubra-Shyok-Indus system; map claims must remain distinct from boundary claims. The other options describe different processes, locations, scales or governance categories.

Q47. Which statement preserves the process boundary for Himalayan map discipline?

A. A GLOF is sudden lake release after overtopping or dam failure; triggers can include slope failure, ice or rock entry, intense inflow, piping, erosion or seismic disturbance. B. GLOF hazard depends on lake, dam, slope and breach processes; risk additionally requires exposed people, settlements, roads, hydropower and limited warning or evacuation capacity. C. Gangotri links to the Bhagirathi, Zemu to the Teesta, Bara Shigri to the Chenab system and Siachen to the Nubra-Shyok-Indus system; map claims must remain distinct from boundary claims. D. South Lhonak was a moraine-dammed lake involved in the October 2023 Teesta disaster; event-specific work examines slope movement, impulse, overtopping and breach without reducing attribution to warming alone.

Answer: C. Explanation: Gangotri links to the Bhagirathi, Zemu to the Teesta, Bara Shigri to the Chenab system and Siachen to the Nubra-Shyok-Indus system; map claims must remain distinct from boundary claims. The other options describe different processes, locations, scales or governance categories.

Q48. Which option avoids the main UPSC trap concerning Himalayan map discipline?

A. South Lhonak was a moraine-dammed lake involved in the October 2023 Teesta disaster; event-specific work examines slope movement, impulse, overtopping and breach without reducing attribution to warming alone. B. Inventory and remote sensing must lead to field assessment, thresholds and warning, communication and drills, land-use controls, evacuation routes and GLOF-informed infrastructure design. C. GLOF hazard depends on lake, dam, slope and breach processes; risk additionally requires exposed people, settlements, roads, hydropower and limited warning or evacuation capacity. D. Gangotri links to the Bhagirathi, Zemu to the Teesta, Bara Shigri to the Chenab system and Siachen to the Nubra-Shyok-Indus system; map claims must remain distinct from boundary claims.

Answer: D. Explanation: Gangotri links to the Bhagirathi, Zemu to the Teesta, Bara Shigri to the Chenab system and Siachen to the Nubra-Shyok-Indus system; map claims must remain distinct from boundary claims. The other options describe different processes, locations, scales or governance categories.

Q49. Which statement correctly explains Water-timing effect?

A. Sustained ice loss can initially raise local melt contribution but later weaken lean-season and drought-year buffering; monsoon rain, snow and groundwater prevent a uniform all-rivers-dry conclusion. B. GLOF hazard depends on lake, dam, slope and breach processes; risk additionally requires exposed people, settlements, roads, hydropower and limited warning or evacuation capacity. C. Retreat can expose overdeepened basins or leave moraine dams that store water; lake existence alone does not establish instability or downstream risk. D. A GLOF is sudden lake release after overtopping or dam failure; triggers can include slope failure, ice or rock entry, intense inflow, piping, erosion or seismic disturbance.

Answer: A. Explanation: Sustained ice loss can initially raise local melt contribution but later weaken lean-season and drought-year buffering; monsoon rain, snow and groundwater prevent a uniform all-rivers-dry conclusion. The other options describe different processes, locations, scales or governance categories.

Q50. Which option is the safest spatial interpretation of Water-timing effect?

A. A GLOF is sudden lake release after overtopping or dam failure; triggers can include slope failure, ice or rock entry, intense inflow, piping, erosion or seismic disturbance. B. Sustained ice loss can initially raise local melt contribution but later weaken lean-season and drought-year buffering; monsoon rain, snow and groundwater prevent a uniform all-rivers-dry conclusion. C. South Lhonak was a moraine-dammed lake involved in the October 2023 Teesta disaster; event-specific work examines slope movement, impulse, overtopping and breach without reducing attribution to warming alone. D. GLOF hazard depends on lake, dam, slope and breach processes; risk additionally requires exposed people, settlements, roads, hydropower and limited warning or evacuation capacity.

Answer: B. Explanation: Sustained ice loss can initially raise local melt contribution but later weaken lean-season and drought-year buffering; monsoon rain, snow and groundwater prevent a uniform all-rivers-dry conclusion. The other options describe different processes, locations, scales or governance categories.

Q51. Which statement preserves the process boundary for Water-timing effect?

A. GLOF hazard depends on lake, dam, slope and breach processes; risk additionally requires exposed people, settlements, roads, hydropower and limited warning or evacuation capacity. B. Inventory and remote sensing must lead to field assessment, thresholds and warning, communication and drills, land-use controls, evacuation routes and GLOF-informed infrastructure design. C. Sustained ice loss can initially raise local melt contribution but later weaken lean-season and drought-year buffering; monsoon rain, snow and groundwater prevent a uniform all-rivers-dry conclusion. D. South Lhonak was a moraine-dammed lake involved in the October 2023 Teesta disaster; event-specific work examines slope movement, impulse, overtopping and breach without reducing attribution to warming alone.

Answer: C. Explanation: Sustained ice loss can initially raise local melt contribution but later weaken lean-season and drought-year buffering; monsoon rain, snow and groundwater prevent a uniform all-rivers-dry conclusion. The other options describe different processes, locations, scales or governance categories.

Q52. Which option avoids the main UPSC trap concerning Water-timing effect?

A. Direct routed demands include 2020 GS-I on glacier melting and Indian water resources, 2023 GS-I on fjords and 2019 Prelims glacier-river matching. B. Inventory and remote sensing must lead to field assessment, thresholds and warning, communication and drills, land-use controls, evacuation routes and GLOF-informed infrastructure design. C. South Lhonak was a moraine-dammed lake involved in the October 2023 Teesta disaster; event-specific work examines slope movement, impulse, overtopping and breach without reducing attribution to warming alone. D. Sustained ice loss can initially raise local melt contribution but later weaken lean-season and drought-year buffering; monsoon rain, snow and groundwater prevent a uniform all-rivers-dry conclusion.

Answer: D. Explanation: Sustained ice loss can initially raise local melt contribution but later weaken lean-season and drought-year buffering; monsoon rain, snow and groundwater prevent a uniform all-rivers-dry conclusion. The other options describe different processes, locations, scales or governance categories.

Q53. Which statement correctly explains Glacial-lake formation?

A. Retreat can expose overdeepened basins or leave moraine dams that store water; lake existence alone does not establish instability or downstream risk. B. GLOF hazard depends on lake, dam, slope and breach processes; risk additionally requires exposed people, settlements, roads, hydropower and limited warning or evacuation capacity. C. A GLOF is sudden lake release after overtopping or dam failure; triggers can include slope failure, ice or rock entry, intense inflow, piping, erosion or seismic disturbance. D. South Lhonak was a moraine-dammed lake involved in the October 2023 Teesta disaster; event-specific work examines slope movement, impulse, overtopping and breach without reducing attribution to warming alone.

Answer: A. Explanation: Retreat can expose overdeepened basins or leave moraine dams that store water; lake existence alone does not establish instability or downstream risk. The other options describe different processes, locations, scales or governance categories.

Q54. Which option is the safest spatial interpretation of Glacial-lake formation?

A. Inventory and remote sensing must lead to field assessment, thresholds and warning, communication and drills, land-use controls, evacuation routes and GLOF-informed infrastructure design. B. Retreat can expose overdeepened basins or leave moraine dams that store water; lake existence alone does not establish instability or downstream risk. C. South Lhonak was a moraine-dammed lake involved in the October 2023 Teesta disaster; event-specific work examines slope movement, impulse, overtopping and breach without reducing attribution to warming alone. D. GLOF hazard depends on lake, dam, slope and breach processes; risk additionally requires exposed people, settlements, roads, hydropower and limited warning or evacuation capacity.

Answer: B. Explanation: Retreat can expose overdeepened basins or leave moraine dams that store water; lake existence alone does not establish instability or downstream risk. The other options describe different processes, locations, scales or governance categories.

Q55. Which statement preserves the process boundary for Glacial-lake formation?

A. Inventory and remote sensing must lead to field assessment, thresholds and warning, communication and drills, land-use controls, evacuation routes and GLOF-informed infrastructure design. B. South Lhonak was a moraine-dammed lake involved in the October 2023 Teesta disaster; event-specific work examines slope movement, impulse, overtopping and breach without reducing attribution to warming alone. C. Retreat can expose overdeepened basins or leave moraine dams that store water; lake existence alone does not establish instability or downstream risk. D. Direct routed demands include 2020 GS-I on glacier melting and Indian water resources, 2023 GS-I on fjords and 2019 Prelims glacier-river matching.

Answer: C. Explanation: Retreat can expose overdeepened basins or leave moraine dams that store water; lake existence alone does not establish instability or downstream risk. The other options describe different processes, locations, scales or governance categories.

Q56. Which option avoids the main UPSC trap concerning Glacial-lake formation?

A. Inventory and remote sensing must lead to field assessment, thresholds and warning, communication and drills, land-use controls, evacuation routes and GLOF-informed infrastructure design. B. ISRO SAC's Himalayan Glacier Inventory Atlas verifies an official mapping platform; it does not by itself prove a current retreat rate, loss percentage, lake ranking or forecast. C. Direct routed demands include 2020 GS-I on glacier melting and Indian water resources, 2023 GS-I on fjords and 2019 Prelims glacier-river matching. D. Retreat can expose overdeepened basins or leave moraine dams that store water; lake existence alone does not establish instability or downstream risk.

Answer: D. Explanation: Retreat can expose overdeepened basins or leave moraine dams that store water; lake existence alone does not establish instability or downstream risk. The other options describe different processes, locations, scales or governance categories.

Q57. Which statement correctly explains GLOF mechanism?

A. A GLOF is sudden lake release after overtopping or dam failure; triggers can include slope failure, ice or rock entry, intense inflow, piping, erosion or seismic disturbance. B. GLOF hazard depends on lake, dam, slope and breach processes; risk additionally requires exposed people, settlements, roads, hydropower and limited warning or evacuation capacity. C. Inventory and remote sensing must lead to field assessment, thresholds and warning, communication and drills, land-use controls, evacuation routes and GLOF-informed infrastructure design. D. South Lhonak was a moraine-dammed lake involved in the October 2023 Teesta disaster; event-specific work examines slope movement, impulse, overtopping and breach without reducing attribution to warming alone.

Answer: A. Explanation: A GLOF is sudden lake release after overtopping or dam failure; triggers can include slope failure, ice or rock entry, intense inflow, piping, erosion or seismic disturbance. The other options describe different processes, locations, scales or governance categories.

Q58. Which option is the safest spatial interpretation of GLOF mechanism?

A. Inventory and remote sensing must lead to field assessment, thresholds and warning, communication and drills, land-use controls, evacuation routes and GLOF-informed infrastructure design. B. A GLOF is sudden lake release after overtopping or dam failure; triggers can include slope failure, ice or rock entry, intense inflow, piping, erosion or seismic disturbance. C. Direct routed demands include 2020 GS-I on glacier melting and Indian water resources, 2023 GS-I on fjords and 2019 Prelims glacier-river matching. D. South Lhonak was a moraine-dammed lake involved in the October 2023 Teesta disaster; event-specific work examines slope movement, impulse, overtopping and breach without reducing attribution to warming alone.

Answer: B. Explanation: A GLOF is sudden lake release after overtopping or dam failure; triggers can include slope failure, ice or rock entry, intense inflow, piping, erosion or seismic disturbance. The other options describe different processes, locations, scales or governance categories.

Q59. Which statement preserves the process boundary for GLOF mechanism?

A. Direct routed demands include 2020 GS-I on glacier melting and Indian water resources, 2023 GS-I on fjords and 2019 Prelims glacier-river matching. B. Inventory and remote sensing must lead to field assessment, thresholds and warning, communication and drills, land-use controls, evacuation routes and GLOF-informed infrastructure design. C. A GLOF is sudden lake release after overtopping or dam failure; triggers can include slope failure, ice or rock entry, intense inflow, piping, erosion or seismic disturbance. D. ISRO SAC's Himalayan Glacier Inventory Atlas verifies an official mapping platform; it does not by itself prove a current retreat rate, loss percentage, lake ranking or forecast.

Answer: C. Explanation: A GLOF is sudden lake release after overtopping or dam failure; triggers can include slope failure, ice or rock entry, intense inflow, piping, erosion or seismic disturbance. The other options describe different processes, locations, scales or governance categories.

Q60. Which option avoids the main UPSC trap concerning GLOF mechanism?

A. ISRO SAC's Himalayan Glacier Inventory Atlas verifies an official mapping platform; it does not by itself prove a current retreat rate, loss percentage, lake ranking or forecast. B. Direct routed demands include 2020 GS-I on glacier melting and Indian water resources, 2023 GS-I on fjords and 2019 Prelims glacier-river matching. C. A glacier is compacted and recrystallised land ice thick enough to deform or slide under gravity; a persistent snowfield need not be a glacier because it may not flow. D. A GLOF is sudden lake release after overtopping or dam failure; triggers can include slope failure, ice or rock entry, intense inflow, piping, erosion or seismic disturbance.

Answer: D. Explanation: A GLOF is sudden lake release after overtopping or dam failure; triggers can include slope failure, ice or rock entry, intense inflow, piping, erosion or seismic disturbance. The other options describe different processes, locations, scales or governance categories.

Q61. Which statement correctly explains Hazard-risk distinction?

A. GLOF hazard depends on lake, dam, slope and breach processes; risk additionally requires exposed people, settlements, roads, hydropower and limited warning or evacuation capacity. B. South Lhonak was a moraine-dammed lake involved in the October 2023 Teesta disaster; event-specific work examines slope movement, impulse, overtopping and breach without reducing attribution to warming alone. C. Direct routed demands include 2020 GS-I on glacier melting and Indian water resources, 2023 GS-I on fjords and 2019 Prelims glacier-river matching. D. Inventory and remote sensing must lead to field assessment, thresholds and warning, communication and drills, land-use controls, evacuation routes and GLOF-informed infrastructure design.

Answer: A. Explanation: GLOF hazard depends on lake, dam, slope and breach processes; risk additionally requires exposed people, settlements, roads, hydropower and limited warning or evacuation capacity. The other options describe different processes, locations, scales or governance categories.

Q62. Which option is the safest spatial interpretation of Hazard-risk distinction?

A. Inventory and remote sensing must lead to field assessment, thresholds and warning, communication and drills, land-use controls, evacuation routes and GLOF-informed infrastructure design. B. GLOF hazard depends on lake, dam, slope and breach processes; risk additionally requires exposed people, settlements, roads, hydropower and limited warning or evacuation capacity. C. ISRO SAC's Himalayan Glacier Inventory Atlas verifies an official mapping platform; it does not by itself prove a current retreat rate, loss percentage, lake ranking or forecast. D. Direct routed demands include 2020 GS-I on glacier melting and Indian water resources, 2023 GS-I on fjords and 2019 Prelims glacier-river matching.

Answer: B. Explanation: GLOF hazard depends on lake, dam, slope and breach processes; risk additionally requires exposed people, settlements, roads, hydropower and limited warning or evacuation capacity. The other options describe different processes, locations, scales or governance categories.

Q63. Which statement preserves the process boundary for Hazard-risk distinction?

A. Direct routed demands include 2020 GS-I on glacier melting and Indian water resources, 2023 GS-I on fjords and 2019 Prelims glacier-river matching. B. ISRO SAC's Himalayan Glacier Inventory Atlas verifies an official mapping platform; it does not by itself prove a current retreat rate, loss percentage, lake ranking or forecast. C. GLOF hazard depends on lake, dam, slope and breach processes; risk additionally requires exposed people, settlements, roads, hydropower and limited warning or evacuation capacity. D. A glacier is compacted and recrystallised land ice thick enough to deform or slide under gravity; a persistent snowfield need not be a glacier because it may not flow.

Answer: C. Explanation: GLOF hazard depends on lake, dam, slope and breach processes; risk additionally requires exposed people, settlements, roads, hydropower and limited warning or evacuation capacity. The other options describe different processes, locations, scales or governance categories.

Q64. Which option avoids the main UPSC trap concerning Hazard-risk distinction?

A. Accumulation adds snow and refrozen water, ablation removes mass, and net balance is their difference over a stated glacier and period. B. ISRO SAC's Himalayan Glacier Inventory Atlas verifies an official mapping platform; it does not by itself prove a current retreat rate, loss percentage, lake ranking or forecast. C. A glacier is compacted and recrystallised land ice thick enough to deform or slide under gravity; a persistent snowfield need not be a glacier because it may not flow. D. GLOF hazard depends on lake, dam, slope and breach processes; risk additionally requires exposed people, settlements, roads, hydropower and limited warning or evacuation capacity.

Answer: D. Explanation: GLOF hazard depends on lake, dam, slope and breach processes; risk additionally requires exposed people, settlements, roads, hydropower and limited warning or evacuation capacity. The other options describe different processes, locations, scales or governance categories.

Q65. Which statement correctly explains South Lhonak boundary?

A. South Lhonak was a moraine-dammed lake involved in the October 2023 Teesta disaster; event-specific work examines slope movement, impulse, overtopping and breach without reducing attribution to warming alone. B. Direct routed demands include 2020 GS-I on glacier melting and Indian water resources, 2023 GS-I on fjords and 2019 Prelims glacier-river matching. C. ISRO SAC's Himalayan Glacier Inventory Atlas verifies an official mapping platform; it does not by itself prove a current retreat rate, loss percentage, lake ranking or forecast. D. Inventory and remote sensing must lead to field assessment, thresholds and warning, communication and drills, land-use controls, evacuation routes and GLOF-informed infrastructure design.

Answer: A. Explanation: South Lhonak was a moraine-dammed lake involved in the October 2023 Teesta disaster; event-specific work examines slope movement, impulse, overtopping and breach without reducing attribution to warming alone. The other options describe different processes, locations, scales or governance categories.

Q66. Which option is the safest spatial interpretation of South Lhonak boundary?

A. Direct routed demands include 2020 GS-I on glacier melting and Indian water resources, 2023 GS-I on fjords and 2019 Prelims glacier-river matching. B. South Lhonak was a moraine-dammed lake involved in the October 2023 Teesta disaster; event-specific work examines slope movement, impulse, overtopping and breach without reducing attribution to warming alone. C. ISRO SAC's Himalayan Glacier Inventory Atlas verifies an official mapping platform; it does not by itself prove a current retreat rate, loss percentage, lake ranking or forecast. D. A glacier is compacted and recrystallised land ice thick enough to deform or slide under gravity; a persistent snowfield need not be a glacier because it may not flow.

Answer: B. Explanation: South Lhonak was a moraine-dammed lake involved in the October 2023 Teesta disaster; event-specific work examines slope movement, impulse, overtopping and breach without reducing attribution to warming alone. The other options describe different processes, locations, scales or governance categories.

Q67. Which statement preserves the process boundary for South Lhonak boundary?

A. A glacier is compacted and recrystallised land ice thick enough to deform or slide under gravity; a persistent snowfield need not be a glacier because it may not flow. B. ISRO SAC's Himalayan Glacier Inventory Atlas verifies an official mapping platform; it does not by itself prove a current retreat rate, loss percentage, lake ranking or forecast. C. South Lhonak was a moraine-dammed lake involved in the October 2023 Teesta disaster; event-specific work examines slope movement, impulse, overtopping and breach without reducing attribution to warming alone. D. Accumulation adds snow and refrozen water, ablation removes mass, and net balance is their difference over a stated glacier and period.

Answer: C. Explanation: South Lhonak was a moraine-dammed lake involved in the October 2023 Teesta disaster; event-specific work examines slope movement, impulse, overtopping and breach without reducing attribution to warming alone. The other options describe different processes, locations, scales or governance categories.

Q68. Which option avoids the main UPSC trap concerning South Lhonak boundary?

A. A glacier is compacted and recrystallised land ice thick enough to deform or slide under gravity; a persistent snowfield need not be a glacier because it may not flow. B. The equilibrium-line altitude balances annual accumulation and ablation; the climatic snowline is a long-term regional limit, while the snout is the glacier terminus. C. Accumulation adds snow and refrozen water, ablation removes mass, and net balance is their difference over a stated glacier and period. D. South Lhonak was a moraine-dammed lake involved in the October 2023 Teesta disaster; event-specific work examines slope movement, impulse, overtopping and breach without reducing attribution to warming alone.

Answer: D. Explanation: South Lhonak was a moraine-dammed lake involved in the October 2023 Teesta disaster; event-specific work examines slope movement, impulse, overtopping and breach without reducing attribution to warming alone. The other options describe different processes, locations, scales or governance categories.

Q69. Which statement correctly explains Risk-reduction stack?

A. Inventory and remote sensing must lead to field assessment, thresholds and warning, communication and drills, land-use controls, evacuation routes and GLOF-informed infrastructure design. B. A glacier is compacted and recrystallised land ice thick enough to deform or slide under gravity; a persistent snowfield need not be a glacier because it may not flow. C. Direct routed demands include 2020 GS-I on glacier melting and Indian water resources, 2023 GS-I on fjords and 2019 Prelims glacier-river matching. D. ISRO SAC's Himalayan Glacier Inventory Atlas verifies an official mapping platform; it does not by itself prove a current retreat rate, loss percentage, lake ranking or forecast.

Answer: A. Explanation: Inventory and remote sensing must lead to field assessment, thresholds and warning, communication and drills, land-use controls, evacuation routes and GLOF-informed infrastructure design. The other options describe different processes, locations, scales or governance categories.

Q70. Which option is the safest spatial interpretation of Risk-reduction stack?

A. A glacier is compacted and recrystallised land ice thick enough to deform or slide under gravity; a persistent snowfield need not be a glacier because it may not flow. B. Inventory and remote sensing must lead to field assessment, thresholds and warning, communication and drills, land-use controls, evacuation routes and GLOF-informed infrastructure design. C. Accumulation adds snow and refrozen water, ablation removes mass, and net balance is their difference over a stated glacier and period. D. ISRO SAC's Himalayan Glacier Inventory Atlas verifies an official mapping platform; it does not by itself prove a current retreat rate, loss percentage, lake ranking or forecast.

Answer: B. Explanation: Inventory and remote sensing must lead to field assessment, thresholds and warning, communication and drills, land-use controls, evacuation routes and GLOF-informed infrastructure design. The other options describe different processes, locations, scales or governance categories.

Q71. Which statement preserves the process boundary for Risk-reduction stack?

A. A glacier is compacted and recrystallised land ice thick enough to deform or slide under gravity; a persistent snowfield need not be a glacier because it may not flow. B. The equilibrium-line altitude balances annual accumulation and ablation; the climatic snowline is a long-term regional limit, while the snout is the glacier terminus. C. Inventory and remote sensing must lead to field assessment, thresholds and warning, communication and drills, land-use controls, evacuation routes and GLOF-informed infrastructure design. D. Accumulation adds snow and refrozen water, ablation removes mass, and net balance is their difference over a stated glacier and period.

Answer: C. Explanation: Inventory and remote sensing must lead to field assessment, thresholds and warning, communication and drills, land-use controls, evacuation routes and GLOF-informed infrastructure design. The other options describe different processes, locations, scales or governance categories.

Q72. Which option avoids the main UPSC trap concerning Risk-reduction stack?

A. Accumulation adds snow and refrozen water, ablation removes mass, and net balance is their difference over a stated glacier and period. B. The equilibrium-line altitude balances annual accumulation and ablation; the climatic snowline is a long-term regional limit, while the snout is the glacier terminus. C. Ice can continue moving down-valley while the snout retreats when ablation at the terminus exceeds forward ice delivery. D. Inventory and remote sensing must lead to field assessment, thresholds and warning, communication and drills, land-use controls, evacuation routes and GLOF-informed infrastructure design.

Answer: D. Explanation: Inventory and remote sensing must lead to field assessment, thresholds and warning, communication and drills, land-use controls, evacuation routes and GLOF-informed infrastructure design. The other options describe different processes, locations, scales or governance categories.

Q73. Which statement correctly explains Verified PYQ routes?

A. Direct routed demands include 2020 GS-I on glacier melting and Indian water resources, 2023 GS-I on fjords and 2019 Prelims glacier-river matching. B. ISRO SAC's Himalayan Glacier Inventory Atlas verifies an official mapping platform; it does not by itself prove a current retreat rate, loss percentage, lake ranking or forecast. C. A glacier is compacted and recrystallised land ice thick enough to deform or slide under gravity; a persistent snowfield need not be a glacier because it may not flow. D. Accumulation adds snow and refrozen water, ablation removes mass, and net balance is their difference over a stated glacier and period.

Answer: A. Explanation: Direct routed demands include 2020 GS-I on glacier melting and Indian water resources, 2023 GS-I on fjords and 2019 Prelims glacier-river matching. The other options describe different processes, locations, scales or governance categories.

Demand decoding: The directive **answer** requires a direct position on "Q73. Which statement correctly explains Verified PYQ routes?", all clauses and scales, a process chain, spatial reconstruction, named India/world evidence, a counter-condition and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: The answer must resolve the physical-geography demand in "Q73. Which statement correctly explains Verified PYQ routes?".

Analytical body:

- Claim and named evidence:** Q73. Which statement correctly explains Verified PYQ routes? **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
- Claim and named evidence:** A. Direct routed demands include 2020 GS-I on glacier melting and Indian water resources, 2023 GS-I on fjords and 2019 Prelims glacier-river matching. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
- Claim and named evidence:** B. ISRO SAC's Himalayan Glacier Inventory Atlas verifies an official mapping platform; it does not by itself prove a current retreat rate, loss percentage, lake ranking or forecast. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or

observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

4. **Claim and named evidence:** C. A glacier is compacted and recrystallised land ice thick enough to deform or slide under gravity; a persistent snowfield need not be a glacier because it may not flow. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

5. **Claim and named evidence:** D. Accumulation adds snow and refrozen water, ablation removes mass, and net balance is their difference over a stated glacier and period. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

Counter-position / limit: A spatial association, one event, one model or one map layer does not establish a sufficient or timeless cause; test energy, material, structure, circulation, scale and human mediation.

Qualified conclusion: The answer must resolve the physical-geography demand in "Q73. Which statement correctly explains Verified PYQ routes?".

Executable exam-length answer / compression plan: For a 15-mark answer, spend about one-sixth of the time decoding and drawing the map/flow; open with definition and thesis; organise five process-spatial points as claim -> named evidence -> analysis -> qualification; compress examples before mechanisms and reserve the final minute for the causal limit.

Why this earns marks: The answer obeys the directive, explains rather than catalogues, makes the spatial logic visible and avoids deterministic or timeless claims.

How to improve this answer: For "Q73. Which statement correctly explains Verified PYQ routes?", replace the weakest generalisation with one labelled process arrow or map anchor and state the scale, exception or evidence needed before extending the conclusion.

Q74. Which option is the safest spatial interpretation of Verified PYQ routes?

A. A glacier is compacted and recrystallised land ice thick enough to deform or slide under gravity; a persistent snowfield need not be a glacier because it may not flow. B. Direct routed demands include 2020 GS-I on glacier melting and Indian water resources, 2023 GS-I on fjords and 2019 Prelims glacier-river matching. C. The equilibrium-line altitude balances annual accumulation and ablation; the climatic snowline is a long-term regional limit, while the snout is the glacier terminus. D. Accumulation adds snow and refrozen water, ablation removes mass, and net balance is their difference over a stated glacier and period.

Answer: B. Explanation: Direct routed demands include 2020 GS-I on glacier melting and Indian water resources, 2023 GS-I on fjords and 2019 Prelims glacier-river matching. The other options describe different processes, locations, scales or governance categories.

Demand decoding: Treat "Q74. Which option is the safest spatial interpretation of Verified PYQ routes?" as a process, location, scale, terminology and status problem. Verify every statement independently.

Detailed examiner-grade model answer:

Introduction and thesis: The answer must resolve the physical-geography demand in "Q74. Which option is the safest spatial interpretation of Verified PYQ routes?".

Analytical body:

1. **Claim and named evidence:** Q74. Which option is the safest spatial interpretation of Verified PYQ routes?
Analysis: Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
2. **Claim and named evidence:** A. A glacier is compacted and recrystallised land ice thick enough to deform or slide under gravity; a persistent snowfield need not be a glacier because it may not flow. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date

boundary.

3. **Claim and named evidence:** B. Direct routed demands include 2020 GS-I on glacier melting and Indian water resources, 2023 GS-I on fjords and 2019 Prelims glacier-river matching. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive.

Qualification: State the scale, threshold, interacting control, exception or source/date boundary.

4. **Claim and named evidence:** C. The equilibrium-line altitude balances annual accumulation and ablation; the climatic snowline is a long-term regional limit, while the snout is the glacier terminus. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

5. **Claim and named evidence:** D. Accumulation adds snow and refrozen water, ablation removes mass, and net balance is their difference over a stated glacier and period. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive.

Qualification: State the scale, threshold, interacting control, exception or source/date boundary.

Counter-position / limit: A spatial association, one event, one model or one map layer does not establish a sufficient or timeless cause; test energy, material, structure, circulation, scale and human mediation.

Qualified conclusion: The answer must resolve the physical-geography demand in "Q74. Which option is the safest spatial interpretation of Verified PYQ routes?".

Executable exam-length answer / compression plan: Fix the phenomenon and spatial setting; trace the controlling mechanism; test direction, sequence, threshold and India/world anchor; eliminate the closest term, map or causation distractor.

Why this earns marks: It preserves exact process-to-pattern reasoning and prevents a familiar place-name or correlation from substituting for causation.

How to improve this answer: For "Q74. Which option is the safest spatial interpretation of Verified PYQ routes?", state why the nearest distractor fails on process, direction, scale, location, terminology, date or official status.

Q75. Which statement preserves the process boundary for Verified PYQ routes?

A. The equilibrium-line altitude balances annual accumulation and ablation; the climatic snowline is a long-term regional limit, while the snout is the glacier terminus. B. Ice can continue moving down-valley while the snout retreats when ablation at the terminus exceeds forward ice delivery. C. Direct routed demands include 2020 GS-I on glacier melting and Indian water resources, 2023 GS-I on fjords and 2019 Prelims glacier-river matching. D. Accumulation adds snow and refrozen water, ablation removes mass, and net balance is their difference over a stated glacier and period.

Answer: C. Explanation: Direct routed demands include 2020 GS-I on glacier melting and Indian water resources, 2023 GS-I on fjords and 2019 Prelims glacier-river matching. The other options describe different processes, locations, scales or governance categories.

Demand decoding: The directive answer requires a direct position on "Q75. Which statement preserves the process boundary for Verified PYQ routes?", all clauses and scales, a process chain, spatial reconstruction, named India/world evidence, a counter-condition and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: The answer must resolve the physical-geography demand in "Q75. Which statement preserves the process boundary for Verified PYQ routes?".

Analytical body:

1. **Claim and named evidence:** Q75. Which statement preserves the process boundary for Verified PYQ routes?
Analysis: Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
2. **Claim and named evidence:** A. The equilibrium-line altitude balances annual accumulation and ablation; the climatic snowline is a long-term regional limit, while the snout is the glacier terminus. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the

directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

3. **Claim and named evidence:** B. Ice can continue moving down-valley while the snout retreats when ablation at the terminus exceeds forward ice delivery. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
4. **Claim and named evidence:** C. Direct routed demands include 2020 GS-I on glacier melting and Indian water resources, 2023 GS-I on fjords and 2019 Prelims glacier-river matching. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
5. **Claim and named evidence:** D. Accumulation adds snow and refrozen water, ablation removes mass, and net balance is their difference over a stated glacier and period. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

Counter-position / limit: A spatial association, one event, one model or one map layer does not establish a sufficient or timeless cause; test energy, material, structure, circulation, scale and human mediation.

Qualified conclusion: The answer must resolve the physical-geography demand in "Q75. Which statement preserves the process boundary for Verified PYQ routes?".

Executable exam-length answer / compression plan: For a 15-mark answer, spend about one-sixth of the time decoding and drawing the map/flow; open with definition and thesis; organise five process-spatial points as claim -> named evidence -> analysis -> qualification; compress examples before mechanisms and reserve the final minute for the causal limit.

Why this earns marks: The answer obeys the directive, explains rather than catalogues, makes the spatial logic visible and avoids deterministic or timeless claims.

How to improve this answer: For "Q75. Which statement preserves the process boundary for Verified PYQ routes?", replace the weakest generalisation with one labelled process arrow or map anchor and state the scale, exception or evidence needed before extending the conclusion.

Q76. Which option avoids the main UPSC trap concerning Verified PYQ routes?

A. Ice can continue moving down-valley while the snout retreats when ablation at the terminus exceeds forward ice delivery. B. Thin debris can enhance melt by lowering albedo, whereas sufficiently thick debris can insulate; debris-covered tongues may thin without rapid visible snout retreat. C. The equilibrium-line altitude balances annual accumulation and ablation; the climatic snowline is a long-term regional limit, while the snout is the glacier terminus. D. Direct routed demands include 2020 GS-I on glacier melting and Indian water resources, 2023 GS-I on fjords and 2019 Prelims glacier-river matching.

Answer: D. Explanation: Direct routed demands include 2020 GS-I on glacier melting and Indian water resources, 2023 GS-I on fjords and 2019 Prelims glacier-river matching. The other options describe different processes, locations, scales or governance categories.

Q77. Which statement correctly explains Current evidence boundary?

A. ISRO SAC's Himalayan Glacier Inventory Atlas verifies an official mapping platform; it does not by itself prove a current retreat rate, loss percentage, lake ranking or forecast. B. The equilibrium-line altitude balances annual accumulation and ablation; the climatic snowline is a long-term regional limit, while the snout is the glacier terminus. C. Accumulation adds snow and refrozen water, ablation removes mass, and net balance is their difference over a stated glacier and period. D. A glacier is compacted and recrystallised land ice thick enough to deform or slide under gravity; a persistent snowfield need not be a glacier because it may not flow.

Answer: A. Explanation: ISRO SAC's Himalayan Glacier Inventory Atlas verifies an official mapping platform; it does not by itself prove a current retreat rate, loss percentage, lake ranking or forecast. The other options describe different processes, locations, scales or governance categories.

Q78. Which option is the safest spatial interpretation of Current evidence boundary?

A. Accumulation adds snow and refrozen water, ablation removes mass, and net balance is their difference over a stated glacier and period. B. ISRO SAC's Himalayan Glacier Inventory Atlas verifies an official mapping platform; it does not by itself prove a current retreat rate, loss percentage, lake ranking or forecast. C. Ice can continue moving down-valley while the snout retreats when ablation at the terminus exceeds forward ice delivery. D. The equilibrium-line altitude balances annual accumulation and ablation; the climatic snowline is a long-term regional limit, while the snout is the glacier terminus.

Answer: B. Explanation: ISRO SAC's Himalayan Glacier Inventory Atlas verifies an official mapping platform; it does not by itself prove a current retreat rate, loss percentage, lake ranking or forecast. The other options describe different processes, locations, scales or governance categories.

Q79. Which statement preserves the process boundary for Current evidence boundary?

A. The equilibrium-line altitude balances annual accumulation and ablation; the climatic snowline is a long-term regional limit, while the snout is the glacier terminus. B. Thin debris can enhance melt by lowering albedo, whereas sufficiently thick debris can insulate; debris-covered tongues may thin without rapid visible snout retreat. C. ISRO SAC's Himalayan Glacier Inventory Atlas verifies an official mapping platform; it does not by itself prove a current retreat rate, loss percentage, lake ranking or forecast. D. Ice can continue moving down-valley while the snout retreats when ablation at the terminus exceeds forward ice delivery.

Answer: C. Explanation: ISRO SAC's Himalayan Glacier Inventory Atlas verifies an official mapping platform; it does not by itself prove a current retreat rate, loss percentage, lake ranking or forecast. The other options describe different processes, locations, scales or governance categories.

Q80. Which option avoids the main UPSC trap concerning Current evidence boundary?

A. Thin debris can enhance melt by lowering albedo, whereas sufficiently thick debris can insulate; debris-covered tongues may thin without rapid visible snout retreat. B. Plucking detaches jointed blocks and abrasion by entrained debris polishes, striates and grinds bedrock, producing rock flour and asymmetric roche moutonnée forms. C. Ice can continue moving down-valley while the snout retreats when ablation at the terminus exceeds forward ice delivery. D. ISRO SAC's Himalayan Glacier Inventory Atlas verifies an official mapping platform; it does not by itself prove a current retreat rate, loss percentage, lake ranking or forecast.

Answer: D. Explanation: ISRO SAC's Himalayan Glacier Inventory Atlas verifies an official mapping platform; it does not by itself prove a current retreat rate, loss percentage, lake ranking or forecast. The other options describe different processes, locations, scales or governance categories.

PYQS AND ANSWER PRACTICE

Demand decoding: Treat "Q76. Which option avoids the main UPSC trap concerning Verified PYQ routes?" as a process, location, scale, terminology and status problem. Verify every statement independently.

Detailed examiner-grade model answer:

Introduction and thesis: The answer must resolve the physical-geography demand in "Q76. Which option avoids the main UPSC trap concerning Verified PYQ routes?".

Analytical body:

- Claim and named evidence:** Q76. Which option avoids the main UPSC trap concerning Verified PYQ routes?
Analysis: Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
- Claim and named evidence:** A. Ice can continue moving down-valley while the snout retreats when ablation at the terminus exceeds forward ice delivery. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

3. **Claim and named evidence:** B. Thin debris can enhance melt by lowering albedo, whereas sufficiently thick debris can insulate; debris-covered tongues may thin without rapid visible snout retreat. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
4. **Claim and named evidence:** C. The equilibrium-line altitude balances annual accumulation and ablation; the climatic snowline is a long-term regional limit, while the snout is the glacier terminus. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
5. **Claim and named evidence:** D. Direct routed demands include 2020 GS-I on glacier melting and Indian water resources, 2023 GS-I on fjords and 2019 Prelims glacier-river matching. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

Counter-position / limit: A spatial association, one event, one model or one map layer does not establish a sufficient or timeless cause; test energy, material, structure, circulation, scale and human mediation.

Qualified conclusion: The answer must resolve the physical-geography demand in "Q76. Which option avoids the main UPSC trap concerning Verified PYQ routes?".

Executable exam-length answer / compression plan: Fix the phenomenon and spatial setting; trace the controlling mechanism; test direction, sequence, threshold and India/world anchor; eliminate the closest term, map or causation distractor.

Why this earns marks: It preserves exact process-to-pattern reasoning and prevents a familiar place-name or correlation from substituting for causation.

How to improve this answer: For "Q76. Which option avoids the main UPSC trap concerning Verified PYQ routes?", state why the nearest distractor fails on process, direction, scale, location, terminology, date or official status.

VERIFIED PYQ OWNERSHIP AUDIT

Verified routing retains the 2020 and 2023 GS-I questions and the 2019 glacier-river objective route. Older objective keys unavailable locally remain explicitly unverified.

OWNER PYQ LEDGER EXTRACTS

Historical PYQ Integration (2018-2023)

Status: Question-level PYQ demand is integrated into this owner. **Provenance:** Audited local official-paper routing ledgers: _PYQ-ROUTING-MAINS-GS1-GS2-ESSAY-2018-2023.md, _PYQ-ROUTING-PRELIMS-2018-2023.md. **Answer-key rule:** The official 2018-2023 Prelims/CSAT keys are not held locally; no option or answer has been inferred.

- **Years represented:** 2019, 2023
- **Paper(s):** GS-I, Prelims GS-I
- **Routed question demands:** 2

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2019	Prelims GS-I	38	Himalayan rivers and associated glacier sources matching	Objective question; official key unavailable locally	Routed; key unavailable locally	Cover the named fact/concept and its likely statement-level distinctions.
2023	GS-I	6	Formation of fjords and their picturesque character	How are they formed and Why · 10 marks · 150 words	Routed to owning topic	Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion.

What this owner must now support

- Himalayan rivers and associated glacier sources matching
- Formation of fjords and their picturesque character

The table integrates the examinable demand and paper metadata. It does not turn an unkeyed objective question into a solved answer, and it does not claim that lexical presence alone proves full conceptual sufficiency.

Historical PYQ Integration (2018-2023)

Status: Question-level PYQ demand is integrated into this owner. **Provenance:** Audited local official-paper routing ledgers: _PYQ-ROUTING-MAINS-GS1-GS2-ESSAY-2018-2023.md.

- **Years represented:** 2020
- **Paper(s):** GS-I
- **Routed question demands:** 1

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2020	GS-I	6	Melting Himalayan glaciers and India's water resources	How will it impact · 10 marks · 150 words	Routed to owning topic	Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion.

What this owner must now support

- Melting Himalayan glaciers and India's water resources

The table integrates the examinable demand and paper metadata. It does not turn an unkeyed objective question into a solved answer, and it does not claim that lexical presence alone proves full conceptual sufficiency.

PYQ DEMAND CARD 1 - 2020 GS-I

Demand: How will the melting of Himalayan glaciers have a far-reaching impact on the water resources of India? (150 words)

Status: Verified routed direct Mains demand.

Model solution: Explain phased water timing: local melt may initially increase, but reduced stored ice can later weaken lean-season buffering. Link irrigation, hydropower, sediment, ecology and hazard, then qualify with monsoon rain, snow, groundwater, debris and basin-specific glacier fractions.

Demand decoding: The directive **answer** requires a direct position on "PYQ DEMAND CARD 1 - 2020 GS-I", all clauses and scales, a process chain, spatial reconstruction, named India/world evidence, a counter-condition and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: The answer must resolve the physical-geography demand in "PYQ DEMAND CARD 1 - 2020 GS-I".

Analytical body:

1. **Claim and named evidence:** Demand: How will the melting of Himalayan glaciers have a far-reaching impact on the water resources of India? (150 words) **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

Counter-position / limit: A spatial association, one event, one model or one map layer does not establish a sufficient or timeless cause; test energy, material, structure, circulation, scale and human mediation.

Qualified conclusion: The answer must resolve the physical-geography demand in "PYQ DEMAND CARD 1 - 2020 GS-I".

Executable exam-length answer / compression plan: For a 15-mark answer, spend about one-sixth of the time decoding and drawing the map/flow; open with definition and thesis; organise five process-spatial points as claim -> named evidence -> analysis -> qualification; compress examples before mechanisms and reserve the final minute for the causal limit.

Why this earns marks: The answer obeys the directive, explains rather than catalogues, makes the spatial logic visible and avoids deterministic or timeless claims.

How to improve this answer: For "PYQ DEMAND CARD 1 - 2020 GS-I", replace the weakest generalisation with one labelled process arrow or map anchor and state the scale, exception or evidence needed before extending the conclusion.

PYQ DEMAND CARD 2 - 2023 GS-I

Demand: How are fjords formed and why do they constitute picturesque areas? (150 words)

Status: Verified routed direct Mains demand.

Model solution: Trace valley-glacier plucking and abrasion to an overdeepened U-trough, then deglaciation and marine inundation. Steep walls, deep water and hanging-valley waterfalls explain scenery. Distinguish a ria as a drowned river valley.

Demand decoding: The directive **answer** requires a direct position on "PYQ DEMAND CARD 2 - 2023 GS-I", all clauses and scales, a process chain, spatial reconstruction, named India/world evidence, a counter-condition and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: The answer must resolve the physical-geography demand in "PYQ DEMAND CARD 2 - 2023 GS-I".

Analytical body:

1. **Claim and named evidence:** Demand: How are fjords formed and why do they constitute picturesque areas? (150 words) **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

Counter-position / limit: A spatial association, one event, one model or one map layer does not establish a sufficient or timeless cause; test energy, material, structure, circulation, scale and human mediation.

Qualified conclusion: The answer must resolve the physical-geography demand in "PYQ DEMAND CARD 2 - 2023 GS-I".

Executable exam-length answer / compression plan: For a 15-mark answer, spend about one-sixth of the time decoding and drawing the map/flow; open with definition and thesis; organise five process-spatial points as claim -> named evidence -> analysis -> qualification; compress examples before mechanisms and reserve the final minute for the causal limit.

Why this earns marks: The answer obeys the directive, explains rather than catalogues, makes the spatial logic visible and avoids deterministic or timeless claims.

How to improve this answer: For "PYQ DEMAND CARD 2 - 2023 GS-I", replace the weakest generalisation with one labelled process arrow or map anchor and state the scale, exception or evidence needed before extending the conclusion.

PYQ DEMAND CARD 3 - 2019 Prelims GS-I

Demand: Matching Himalayan glaciers with associated river systems.

Status: Verified routed objective demand; official key unavailable locally.

Model solution: Use the secure pairs Gangotri-Bhagirathi, Zemu-Teesta, Bara Shigri-Chenab and Siachen-Nubra. Preserve the official options and label any answer inference separately.

Demand decoding: Treat "PYQ DEMAND CARD 3 - 2019 Prelims GS-I" as a process, location, scale, terminology and status problem. Verify every statement independently.

Detailed examiner-grade model answer:

Introduction and thesis: The answer must resolve the physical-geography demand in "PYQ DEMAND CARD 3 - 2019 Prelims GS-I".

Analytical body:

1. **Claim and named evidence:** Demand: Matching Himalayan glaciers with associated river systems. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
2. **Claim and named evidence:** Status: Verified routed objective demand; official key unavailable locally. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
3. **Claim and named evidence:** Model solution: Use the secure pairs Gangotri-Bhagirathi, Zemu-Teesta, Bara Shigri-Chenab and Siachen-Nubra. Preserve the official options and label any answer inference separately. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

Counter-position / limit: A spatial association, one event, one model or one map layer does not establish a sufficient or timeless cause; test energy, material, structure, circulation, scale and human mediation.

Qualified conclusion: The answer must resolve the physical-geography demand in "PYQ DEMAND CARD 3 - 2019 Prelims GS-I".

Executable exam-length answer / compression plan: Fix the phenomenon and spatial setting; trace the controlling mechanism; test direction, sequence, threshold and India/world anchor; eliminate the closest term, map or causation distractor.

Why this earns marks: It preserves exact process-to-pattern reasoning and prevents a familiar place-name or correlation from substituting for causation.

How to improve this answer: For "PYQ DEMAND CARD 3 - 2019 Prelims GS-I", state why the nearest distractor fails on process, direction, scale, location, terminology, date or official status.

ORIGINAL MAINS 1 - 10 MARKS

Question: Explain glacier mass balance and the apparent paradox of a retreating yet forward-flowing glacier. Answer in about 150 words.

Model thesis: Mass balance controls geometry, while the terminus retreats when local loss exceeds ice delivery even though ice motion remains downslope.

Claim -> named evidence -> analysis -> qualification:

- A glacier is compacted and recrystallised land ice thick enough to deform or slide under gravity; a persistent snowfield need not be a glacier because it may not flow.
- Accumulation adds snow and refrozen water, ablation removes mass, and net balance is their difference over a stated glacier and period.

- The equilibrium-line altitude balances annual accumulation and ablation; the climatic snowline is a long-term regional limit, while the snout is the glacier terminus.
- Ice can continue moving down-valley while the snout retreats when ablation at the terminus exceeds forward ice delivery.
- Thin debris can enhance melt by lowering albedo, whereas sufficiently thick debris can insulate; debris-covered tongues may thin without rapid visible snout retreat.

Qualified conclusion: Mass balance controls geometry, while the terminus retreats when local loss exceeds ice delivery even though ice motion remains downslope.

Demand decoding: The directive **explain** requires a direct position on "Explain glacier mass balance and the apparent paradox of a retreating yet forward-flowing...", all clauses and scales, a process chain, spatial reconstruction, named India/world evidence, a counter-condition and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: Mass balance controls geometry, while the terminus retreats when local loss exceeds ice delivery even though ice motion remains downslope.

Analytical body:

1. **Claim and named evidence:** A glacier is compacted and recrystallised land ice thick enough to deform or slide under gravity; a persistent snowfield need not be a glacier because it may not flow. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
2. **Claim and named evidence:** Accumulation adds snow and refrozen water, ablation removes mass, and net balance is their difference over a stated glacier and period. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
3. **Claim and named evidence:** The equilibrium-line altitude balances annual accumulation and ablation; the climatic snowline is a long-term regional limit, while the snout is the glacier terminus. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
4. **Claim and named evidence:** Ice can continue moving down-valley while the snout retreats when ablation at the terminus exceeds forward ice delivery. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
5. **Claim and named evidence:** Thin debris can enhance melt by lowering albedo, whereas sufficiently thick debris can insulate; debris-covered tongues may thin without rapid visible snout retreat. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

Counter-position / limit: A spatial association, one event, one model or one map layer does not establish a sufficient or timeless cause; test energy, material, structure, circulation, scale and human mediation.

Qualified conclusion: Mass balance controls geometry, while the terminus retreats when local loss exceeds ice delivery even though ice motion remains downslope.

Executable exam-length answer / compression plan: For a 10-mark answer, spend about one-sixth of the time decoding and drawing the map/flow; open with definition and thesis; organise three process-spatial points as claim -> named evidence -> analysis -> qualification; compress examples before mechanisms and reserve the final minute for the causal limit.

Why this earns marks: The answer obeys the directive, explains rather than catalogues, makes the spatial logic visible and avoids deterministic or timeless claims.

How to improve this answer: For "Explain glacier mass balance and the apparent paradox of a retreating yet forward-flowing...", replace the weakest generalisation with one labelled process arrow or map anchor and state the scale, exception or evidence needed before extending the conclusion.

ORIGINAL MAINS 2 - 10 MARKS

Question: Differentiate the principal erosional and depositional landforms of valley glaciation. Answer in about 150 words.

Model thesis: Agent, process, form and sediment sorting distinguish plucking-abrasion landforms from till and meltwater deposits.

Claim -> named evidence -> analysis -> qualification:

- Plucking detaches jointed blocks and abrasion by entrained debris polishes, striates and grinds bedrock, producing rock flour and asymmetric roche moutonnée forms.
- Cirques, aretes and horns develop around glacier heads; valley ice widens and deepens V-valleys into U-shaped troughs and leaves hanging tributary valleys.
- Direct ice deposition is generally unsorted till; lateral, medial, terminal and ground moraines describe debris position or form.
- Eskers are sinuous ridges deposited by subglacial or englacial streams, while outwash plains contain relatively sorted sediment spread by braided meltwater.

Qualified conclusion: Agent, process, form and sediment sorting distinguish plucking-abrasion landforms from till and meltwater deposits.

Demand decoding: The directive **answer** requires a direct position on "Differentiate the principal erosional and depositional landforms of valley glaciation. Answer...", all clauses and scales, a process chain, spatial reconstruction, named India/world evidence, a counter-condition and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: Agent, process, form and sediment sorting distinguish plucking-abrasion landforms from till and meltwater deposits.

Analytical body:

1. **Claim and named evidence:** Plucking detaches jointed blocks and abrasion by entrained debris polishes, striates and grinds bedrock, producing rock flour and asymmetric roche moutonnée forms. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
2. **Claim and named evidence:** Cirques, aretes and horns develop around glacier heads; valley ice widens and deepens V-valleys into U-shaped troughs and leaves hanging tributary valleys. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
3. **Claim and named evidence:** Direct ice deposition is generally unsorted till; lateral, medial, terminal and ground moraines describe debris position or form. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
4. **Claim and named evidence:** Eskers are sinuous ridges deposited by subglacial or englacial streams, while outwash plains contain relatively sorted sediment spread by braided meltwater. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

Counter-position / limit: A spatial association, one event, one model or one map layer does not establish a sufficient or timeless cause; test energy, material, structure, circulation, scale and human mediation.

Qualified conclusion: Agent, process, form and sediment sorting distinguish plucking-abrasion landforms from till and meltwater deposits.

Executable exam-length answer / compression plan: For a 10-mark answer, spend about one-sixth of the time decoding and drawing the map/flow; open with definition and thesis; organise three process-spatial points as claim -> named evidence -> analysis -> qualification; compress examples before mechanisms and reserve the final minute for the causal limit.

Why this earns marks: The answer obeys the directive, explains rather than catalogues, makes the spatial logic visible and avoids deterministic or timeless claims.

How to improve this answer: For "Differentiate the principal erosional and depositional landforms of valley glaciation. Answer...", replace the weakest generalisation with one labelled process arrow or map anchor and state the scale, exception or evidence needed before extending the conclusion.

ORIGINAL MAINS 3 - 15 MARKS

Question: Explain fjord formation and distinguish fjords from rias. Answer in about 250 words.

Model thesis: Valley-glacier overdeepening followed by marine inundation creates fjords; drowned fluvial valleys create rias.

Claim -> named evidence -> analysis -> qualification:

- Cirques, aretes and horns develop around glacier heads; valley ice widens and deepens V-valleys into U-shaped troughs and leaves hanging tributary valleys.
- A fjord is a marine-flooded glacial trough, whereas a ria is a drowned river valley; marine inundation follows rather than creates the glacial excavation.

Qualified conclusion: Valley-glacier overdeepening followed by marine inundation creates fjords; drowned fluvial valleys create rias.

Demand decoding: The directive **explain** requires a direct position on "Explain fjord formation and distinguish fjords from rias. Answer in about 250 words.", all clauses and scales, a process chain, spatial reconstruction, named India/world evidence, a counter-condition and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: Valley-glacier overdeepening followed by marine inundation creates fjords; drowned fluvial valleys create rias.

Analytical body:

1. **Claim and named evidence:** Cirques, aretes and horns develop around glacier heads; valley ice widens and deepens V-valleys into U-shaped troughs and leaves hanging tributary valleys. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
2. **Claim and named evidence:** A fjord is a marine-flooded glacial trough, whereas a ria is a drowned river valley; marine inundation follows rather than creates the glacial excavation. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

Counter-position / limit: A spatial association, one event, one model or one map layer does not establish a sufficient or timeless cause; test energy, material, structure, circulation, scale and human mediation.

Qualified conclusion: Valley-glacier overdeepening followed by marine inundation creates fjords; drowned fluvial valleys create rias.

Executable exam-length answer / compression plan: For a 15-mark answer, spend about one-sixth of the time decoding and drawing the map/flow; open with definition and thesis; organise five process-spatial points as claim -> named evidence -> analysis -> qualification; compress examples before mechanisms and reserve the final minute for the causal limit.

Why this earns marks: The answer obeys the directive, explains rather than catalogues, makes the spatial logic visible and avoids deterministic or timeless claims.

How to improve this answer: For "Explain fjord formation and distinguish fjords from rias. Answer in about 250 words.", replace the weakest generalisation with one labelled process arrow or map anchor and state the scale, exception or evidence needed before extending the conclusion.

ORIGINAL MAINS 4 - 15 MARKS

Question: Assess the far-reaching effects of Himalayan glacier loss on India's water resources. Answer in about 250 words.

Model thesis: The strongest effect is altered timing and reliability in glacierised headwaters, qualified by monsoon, snow, groundwater and basin heterogeneity.

Claim -> named evidence -> analysis -> qualification:

- Gangotri links to the Bhagirathi, Zemu to the Teesta, Bara Shigri to the Chenab system and Siachen to the Nubra-Shyok-Indus system; map claims must remain distinct from boundary claims.
- Sustained ice loss can initially raise local melt contribution but later weaken lean-season and drought-year buffering; monsoon rain, snow and groundwater prevent a uniform all-rivers-dry conclusion.
- ISRO SAC's Himalayan Glacier Inventory Atlas verifies an official mapping platform; it does not by itself prove a current retreat rate, loss percentage, lake ranking or forecast.

Qualified conclusion: The strongest effect is altered timing and reliability in glacierised headwaters, qualified by monsoon, snow, groundwater and basin heterogeneity.

Demand decoding: The directive **assess** requires a direct position on "Assess the far-reaching effects of Himalayan glacier loss on India's water resources. Answer...", all clauses and scales, a process chain, spatial reconstruction, named India/world evidence, a counter-condition and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: The strongest effect is altered timing and reliability in glacierised headwaters, qualified by monsoon, snow, groundwater and basin heterogeneity.

Analytical body:

1. **Claim and named evidence:** Gangotri links to the Bhagirathi, Zemu to the Teesta, Bara Shigri to the Chenab system and Siachen to the Nubra-Shyok-Indus system; map claims must remain distinct from boundary claims. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
2. **Claim and named evidence:** Sustained ice loss can initially raise local melt contribution but later weaken lean-season and drought-year buffering; monsoon rain, snow and groundwater prevent a uniform all-rivers-dry conclusion. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
3. **Claim and named evidence:** ISRO SAC's Himalayan Glacier Inventory Atlas verifies an official mapping platform; it does not by itself prove a current retreat rate, loss percentage, lake ranking or forecast. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

Counter-position / limit: A spatial association, one event, one model or one map layer does not establish a sufficient or timeless cause; test energy, material, structure, circulation, scale and human mediation.

Qualified conclusion: The strongest effect is altered timing and reliability in glacierised headwaters, qualified by monsoon, snow, groundwater and basin heterogeneity.

Executable exam-length answer / compression plan: For a 15-mark answer, spend about one-sixth of the time decoding and drawing the map/flow; open with definition and thesis; organise five process-spatial points as claim -> named evidence -> analysis -> qualification; compress examples before mechanisms and reserve the final minute for the causal limit.

Why this earns marks: The answer obeys the directive, explains rather than catalogues, makes the spatial logic visible and avoids deterministic or timeless claims.

How to improve this answer: For "Assess the far-reaching effects of Himalayan glacier loss on India's water resources. Answer...", replace the weakest generalisation with one labelled process arrow or map anchor and state

the scale, exception or evidence needed before extending the conclusion.

ORIGINAL MAINS 5 - 20 MARKS

Question: Analyse GLOF risk as an interaction of cryospheric change, geomorphic instability and valley-floor development. Answer in about 300 words.

Model thesis: Risk arises from lake-dam-slope hazard plus downstream exposure and vulnerability, not from lake growth or warming alone.

Claim -> named evidence -> analysis -> qualification:

- Retreat can expose overdeepened basins or leave moraine dams that store water; lake existence alone does not establish instability or downstream risk.
- A GLOF is sudden lake release after overtopping or dam failure; triggers can include slope failure, ice or rock entry, intense inflow, piping, erosion or seismic disturbance.
- GLOF hazard depends on lake, dam, slope and breach processes; risk additionally requires exposed people, settlements, roads, hydropower and limited warning or evacuation capacity.
- South Lhonak was a moraine-dammed lake involved in the October 2023 Teesta disaster; event-specific work examines slope movement, impulse, overtopping and breach without reducing attribution to warming alone.
- Inventory and remote sensing must lead to field assessment, thresholds and warning, communication and drills, land-use controls, evacuation routes and GLOF-informed infrastructure design.

Qualified conclusion: Risk arises from lake-dam-slope hazard plus downstream exposure and vulnerability, not from lake growth or warming alone.

Demand decoding: The directive **analyse** requires a direct position on "Analyse GLOF risk as an interaction of cryospheric change, geomorphic instability and valley-...", all clauses and scales, a process chain, spatial reconstruction, named India/world evidence, a counter-condition and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: Risk arises from lake-dam-slope hazard plus downstream exposure and vulnerability, not from lake growth or warming alone.

Analytical body:

1. **Claim and named evidence:** Retreat can expose overdeepened basins or leave moraine dams that store water; lake existence alone does not establish instability or downstream risk. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
2. **Claim and named evidence:** A GLOF is sudden lake release after overtopping or dam failure; triggers can include slope failure, ice or rock entry, intense inflow, piping, erosion or seismic disturbance. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
3. **Claim and named evidence:** GLOF hazard depends on lake, dam, slope and breach processes; risk additionally requires exposed people, settlements, roads, hydropower and limited warning or evacuation capacity. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
4. **Claim and named evidence:** South Lhonak was a moraine-dammed lake involved in the October 2023 Teesta disaster; event-specific work examines slope movement, impulse, overtopping and breach without reducing attribution to warming alone. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
5. **Claim and named evidence:** Inventory and remote sensing must lead to field assessment, thresholds and warning, communication and drills, land-use controls, evacuation routes and GLOF-informed infrastructure design. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named

place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

Counter-position / limit: A spatial association, one event, one model or one map layer does not establish a sufficient or timeless cause; test energy, material, structure, circulation, scale and human mediation.

Qualified conclusion: Risk arises from lake-dam-slope hazard plus downstream exposure and vulnerability, not from lake growth or warming alone.

Executable exam-length answer / compression plan: For a 20-mark answer, spend about one-sixth of the time decoding and drawing the map/flow; open with definition and thesis; organise six to eight process-spatial points as claim -> named evidence -> analysis -> qualification; compress examples before mechanisms and reserve the final minute for the causal limit.

Why this earns marks: The answer obeys the directive, explains rather than catalogues, makes the spatial logic visible and avoids deterministic or timeless claims.

How to improve this answer: For "Analyse GLOF risk as an interaction of cryospheric change, geomorphic instability and valley-...", replace the weakest generalisation with one labelled process arrow or map anchor and state the scale, exception or evidence needed before extending the conclusion.

ORIGINAL MAINS 6 - 20 MARKS

Question: Design a Himalayan cryosphere risk-governance strategy integrating observation, warning, land use and infrastructure. Answer in about 300 words.

Model thesis: A complete strategy joins remote inventory to field thresholds, interoperable warning, evacuation, zoning, dam safety and transparent uncertainty.

Claim -> named evidence -> analysis -> qualification:

- GLOF hazard depends on lake, dam, slope and breach processes; risk additionally requires exposed people, settlements, roads, hydropower and limited warning or evacuation capacity.
- Inventory and remote sensing must lead to field assessment, thresholds and warning, communication and drills, land-use controls, evacuation routes and GLOF-informed infrastructure design.
- ISRO SAC's Himalayan Glacier Inventory Atlas verifies an official mapping platform; it does not by itself prove a current retreat rate, loss percentage, lake ranking or forecast.

Qualified conclusion: A complete strategy joins remote inventory to field thresholds, interoperable warning, evacuation, zoning, dam safety and transparent uncertainty.

Demand decoding: The directive **answer** requires a direct position on "Design a Himalayan cryosphere risk-governance strategy integrating observation, warning, land...", all clauses and scales, a process chain, spatial reconstruction, named India/world evidence, a counter-condition and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: A complete strategy joins remote inventory to field thresholds, interoperable warning, evacuation, zoning, dam safety and transparent uncertainty.

Analytical body:

1. **Claim and named evidence:** GLOF hazard depends on lake, dam, slope and breach processes; risk additionally requires exposed people, settlements, roads, hydropower and limited warning or evacuation capacity. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.
2. **Claim and named evidence:** Inventory and remote sensing must lead to field assessment, thresholds and warning, communication and drills, land-use controls, evacuation routes and GLOF-informed infrastructure design. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

3. **Claim and named evidence:** ISRO SAC's Himalayan Glacier Inventory Atlas verifies an official mapping platform; it does not by itself prove a current retreat rate, loss percentage, lake ranking or forecast. **Analysis:** Trace the driver -> mechanism -> spatial expression -> consequence chain and connect the named place, map or observation to the directive. **Qualification:** State the scale, threshold, interacting control, exception or source/date boundary.

Counter-position / limit: A spatial association, one event, one model or one map layer does not establish a sufficient or timeless cause; test energy, material, structure, circulation, scale and human mediation.

Qualified conclusion: A complete strategy joins remote inventory to field thresholds, interoperable warning, evacuation, zoning, dam safety and transparent uncertainty.

Executable exam-length answer / compression plan: For a 20-mark answer, spend about one-sixth of the time decoding and drawing the map/flow; open with definition and thesis; organise six to eight process-spatial points as claim -> named evidence -> analysis -> qualification; compress examples before mechanisms and reserve the final minute for the causal limit.

Why this earns marks: The answer obeys the directive, explains rather than catalogues, makes the spatial logic visible and avoids deterministic or timeless claims.

How to improve this answer: For "Design a Himalayan cryosphere risk-governance strategy integrating observation, warning, land...", replace the weakest generalisation with one labelled process arrow or map anchor and state the scale, exception or evidence needed before extending the conclusion.

OPTIONAL ADVANCED DEPTH - NOT REQUIRED FOR A CORE ANSWER

Subject: Geography · **Tier:** Advanced (India-applied) · **GS Paper:** GS-I **Grounded in:** D.R. Khullar, *India: A Comprehensive Geography* + Majid Husain, *India geography* + verified CA. **FACT:** = from source book · **ANALYSIS:** = inference / standard knowledge · **CURRENT:** = current affairs. *Companion:* basic/06_Landforms-of-Glaciation.md.

1. Snapshot & core idea

Advanced - Himalayan Glaciers & the Snowline (Khullar / Majid).

FACT: Khullar gives a **source-era estimate** of about 15,000 Himalayan glaciers between the syntaxial bends; modern inventories vary by basin, boundary and minimum mapped size. The snowline is not fixed-it depends on latitude, precipitation, aspect and topography. Most large Indian glaciers lie in the Greater Himalaya and Trans-Himalaya (Karakoram, Ladakh); the Zaskar Range also contains major glaciers. ancient moraines below present snouts prove a Pleistocene glacial age once reached far lower.

- **FACT:** Khullar: about 15,000 glaciers in the Himalaya, between the two syntaxial bends (east & west).
- **FACT:** Snowline = lowest level of perpetual snow; it varies with latitude, precipitation and local topography.
- **FACT:** Greater snowfall/accumulation can lower the climatic snowline, while aridity raises it; aspect, latitude, debris cover and local relief also matter.

- **FACT:** Main glaciers lie in the Greater and Trans-Himalaya (Karakoram, Ladakh, Zaskar); the Lesser Himalaya has only small glaciers (traces in Pir-Panjal, Dhauladhar).
- **FACT:** Siachen lies in the eastern Karakoram under Indian control. Biafo, Baltoro and Hispar are in the Karakoram region administered by Pakistan; they are **regional map references, not Indian glaciers**.

Ca Anchor - South Lhonak Lake & the Teesta GLOF (Sikkim).

FACT: GLOFs are the disaster face of glacial deposition: a moraine-dammed glacial lake that swells with meltwater can breach suddenly, releasing a catastrophic flood.

- **FACT:** South Lhonak is a moraine-dammed glacial lake that expanded as its glacier retreated; use a named satellite study when quoting lake area.
- **FACT:** The October 2023 flood involved lake outburst/overtopping after intense precipitation and a high-mountain mass movement; the exact trigger chain remains a subject of post-event research.

- FACT: Mitigation: glacial-lake monitoring, early-warning systems, and restricting development in the flood path (NDMA GLOF guidelines).

2. Key classification / data

Region / feature	Book-grounded detail	Significance
FACT: Himalayan glaciers	About 15,000 glaciers between eastern and western syntaxial bends in source note	India's frozen water tower
FACT: Snowline	Lowest limit of perpetual snow; varies with latitude, precipitation and topography	Not the same as glacier snout
FACT: Moisture control	Greater accumulation tends to lower snowline; a source-era transect gives a higher dry-Tibetan snowline	Exact difference varies by sector
FACT: Kashmir Himalaya	Some glacier snouts descend lower than in drier inner ranges	Snout altitude is not the same as snowline
FACT: Kumaon / Lahul	Glaciers descend to ~3,660 m in source note	Central Himalayan contrast
FACT: Karakoram	High, arid snowline; Siachen plus Biafo/Baltoro/Hispar as regional map references	Political location must be stated accurately
FACT: GLOF monitoring	Review lake level, overflow paths, moraine condition and downstream exposure	Book hazard-reduction logic plus modern remote sensing/early warning

3. Study links

Study link: FACT: Geography -> India -> Glaciers & Snowline (Khullar: Glaciers of the Himalaya; Majid: Physiography) **Study link:** FACT: Geography -> Glaciation -> Glacial-lake Hazards / GLOF (applied CA)

4. Must-Know Facts (Prelims)

Advanced - Himalayan Glaciers & the Snowline (Khullar / Majid).

- FACT: Khullar's ~15,000 figure is a source-era Himalayan estimate; modern inventory counts are method-dependent.
- FACT: Snowline controls: latitude, precipitation, topography.
- FACT: More accumulation generally lowers snowline; exact north-south contrast varies by Himalayan sector.
- FACT: Karakoram map distinction: Siachen is under Indian control; Biafo/Baltoro/Hispar lie in

Pakistan-administered territory.

- FACT: Do not substitute a glacier-snout elevation for the climatic snowline.

Ca Anchor - South Lhonak Lake & the Teesta GLOF (Sikkim).

- FACT: GLOF = sudden breach of a moraine-dammed glacial lake.
- FACT: South Lhonak (North Sikkim) expanded markedly before the October 2023 GLOF; area estimates are study-specific.
- FACT: Oct 2023 outburst hit the Teesta & Teesta-III dam.

5. UPSC Traps

MEMORY: Trap: Do not confuse similar-looking landforms, processes, scales or Indian locations; UPSC often tests the exact mechanism and setting.

Advanced - Himalayan Glaciers & the Snowline (Khullar / Majid).

- **WRONG:** The snowline is the same as the lowest glacier snout. -> Snowline is the equilibrium/permanent-snow boundary; a glacier tongue may flow far below it.

- WRONG: Snowline and glacier snout are the same thing. -> Snowline = limit of perpetual snow; snout = the glacier's lower terminus (they differ).
- WRONG: Most Indian glaciers are in the Lesser Himalaya. -> Most large glaciers are in the GREATER and TRANS-Himalaya (Karakoram etc.).

Ca Anchor - South Lhonak Lake & the Teesta GLOF (Sikkim).

- WRONG: GLOFs are caused only by earthquakes. -> They result from a moraine-dam breach - often triggered by rain/cloudburst or lake overtopping, not necessarily quakes.

6. CURRENT: Current link

Advanced - Himalayan Glaciers & the Snowline (Khullar / Majid). CURRENT: Modern glacier inventories and mass-balance studies-not one old glacier count-should anchor current cryosphere answers. The Himalaya's snowline and glacier response vary sharply by basin, debris cover and precipitation regime.

Ca Anchor - South Lhonak Lake & the Teesta GLOF (Sikkim). CURRENT: In October 2023 a Glacial Lake Outburst Flood (GLOF) from South Lhonak Lake in North Sikkim surged down the Teesta, destroying infrastructure and damaging the Teesta-III dam.

7. Mains angles

Advanced - Himalayan Glaciers & the Snowline (Khullar / Majid).

- ANALYSIS: Explain why the Himalayan snowline varies with moisture and latitude, and assess the strategic and water-security importance of India's Karakoram glaciers.

Ca Anchor - South Lhonak Lake & the Teesta GLOF (Sikkim).

- ANALYSIS: Link glacial deposition (moraine dams) to GLOF risk and the case for glacial-lake early warning.

Historical PYQ Integration (2018-2023)

Status: Question-level PYQ demand is integrated into this owner. **Provenance:** Audited local official-paper routing ledgers: _PYQ-ROUTING-MAINS-GS1-GS2-ESSAY-2018-2023.md.

- **Years represented:** 2020
- **Paper(s):** GS-I
- **Routed question demands:** 1

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2020	GS-I	6	Melting Himalayan glaciers and India's water resources	How will it impact · 10 marks · 150 words	Routed to owning topic	Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion.

What this owner must now support

- Melting Himalayan glaciers and India's water resources

The table integrates the examinable demand and paper metadata. It does not turn an unkeyed objective question into a solved answer, and it does not claim that lexical presence alone proves full conceptual sufficiency.

CONSOLIDATED REGISTER NOTES

COMPLETE TOPIC ASCII MASTER FLOW DIAGRAM

ASCII MASTER FLOW - PANEL 1/12: Glacier budget

ACCUMULATION -> snowfall, avalanche and refreezing
ELA -> annual gain equals annual loss
ABLATION -> melt, sublimation and calving where relevant
NET BALANCE -> geometry responds with a lag
MUST REMEMBER: Glaciation proceeds through accumulation-ablation balance, ice deformation/basal motion, erosion, transport and deposition; cirque-arête-horn, U-shaped valley, moraine and outwash logic must connect to Himalayan mass balance, proglacial lakes and GLOF cascades.
MUST REMEMBER: Glaciation proceeds through accumulation-ablation balance, ice deformation/basal motion, erosion, transport and deposition; cirque-arête-horn, U-shaped valley, moraine and outwash logic must connect to Himalayan mass balance, proglacial lakes and GLOF cascades.

ASCII MASTER FLOW - PANEL 2/12: Retreat paradox

ICE PARTICLES -> continue moving downslope
TERMINUS -> receives forward ice flux
ABLATION > DELIVERY -> snout shifts up-valley
METRIC RULE -> flow direction differs from margin change

ASCII MASTER FLOW - PANEL 3/12: Erosion suite

PLUCKING -> jointed blocks removed
ABRASION -> polish, striae and rock flour
CIRQUE / ARETE / HORN -> headwall system
U-TROUGH / HANGING VALLEY -> valley-scale overdeepening

ASCII MASTER FLOW - PANEL 4/12: Fjord-ria firewall

FJORD -> glacial trough later flooded by sea
RIA -> river valley later drowned by sea
FJORD CUES -> deep water, steep walls, hanging valleys
RULE -> inundation preserves different pre-existing valleys

ASCII MASTER FLOW - PANEL 5/12: Depositional sorting

TILL / MORaine -> direct ice, generally unsorted
ESKER -> meltwater ridge in or under ice
OUTWASH -> braided meltwater, relatively sorted
DRUMLIN -> streamlined drift/till landform

ASCII MASTER FLOW - PANEL 6/12: Ice-mass family

GROUNDING -> glacier, ice cap, ice sheet
FLOATING -> shelf, iceberg and sea ice
SEA LEVEL -> grounded loss adds ocean mass
BUTTRESSING -> shelf loss can accelerate inland discharge

ASCII MASTER FLOW - PANEL 7/12: Himalayan map hooks

GANGOTRI -> Bhagirathi
ZEMU -> Teesta
BARA SHIGRI -> Chenab system
SIACHEN -> Nubra-Shyok-Indus system

ASCII MASTER FLOW - PANEL 8/12: Water-timing sequence

EARLY LOSS -> local melt contribution may rise
STORED ICE DECLINES -> buffering weakens
LEAN SEASON -> reliability risk grows locally
QUALIFIER -> monsoon, snow and groundwater vary by basin

ASCII MASTER FLOW - PANEL 9/12: Glacial-lake test

BASIN / DAM -> how water is impounded
SLOPE + ICE -> possible displacement trigger
DAM CONDITION -> overtopping, piping or erosion
DOWNSTREAM -> exposure converts hazard into risk

ASCII MASTER FLOW - PANEL 10/12: GLOF cascade

TRIGGER -> slope failure, inflow or instability
WAVE / OVERTOPPING -> dam erosion begins
BREACH -> sudden release + sediment
VALLEY -> constrictions and assets shape consequences
CLOSE DISTINCTION: Snowline, equilibrium-line altitude, firn line and glacier terminus are distinct; till is unsorted ice-laid sediment while outwash is meltwater-sorted, and a fjord is a drowned glacial trough rather than any steep inlet.
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ASCII MASTER FLOW - PANEL 11/12: Risk-reduction stack

SATELLITE INVENTORY -> changing lakes and slopes
FIELD ASSESSMENT -> dam material and thresholds
WARNING + DRILLS -> last-mile action
LAND USE + DESIGN -> reduce exposure and fragility
EVIDENCE LIMIT: Retreat varies by glacier and period; lake growth alone does not prove imminent outburst, while dam material, freeboard, slope instability, ice/rock avalanches and downstream exposure condition risk.
EVIDENCE LIMIT: Retreat varies by glacier and period; lake growth alone does not prove imminent outburst, while dam material, freeboard, slope instability, ice/rock avalanches and downstream exposure condition risk.

ASCII MASTER FLOW - PANEL 12/12: Cryosphere answer spine

DEFINE -> glacier, lake, GLOF and risk
MAP -> glacier-basin relation
EXPLAIN -> budget, trigger and breach
QUALIFY -> heterogeneity, attribution and data date